

Microwave Engineering

EC 401

DoECE

Sardar Vallabhbhai National
Institute Of Technology

Module 2

Rectangular and Circular Wave Guides

Rectangular Wave Guides

WAVEGUIDE INTRODUCTION

- In general terms, a waveguiding system is a system that confines electromagnetic energy and channels it from one point to another.
- This is opposed to a wireless system that uses antennas.

Examples

- Coax
- Twin lead (twisted pair)
- Printed circuit lines (e.g. microstrip)

Transmission Lines

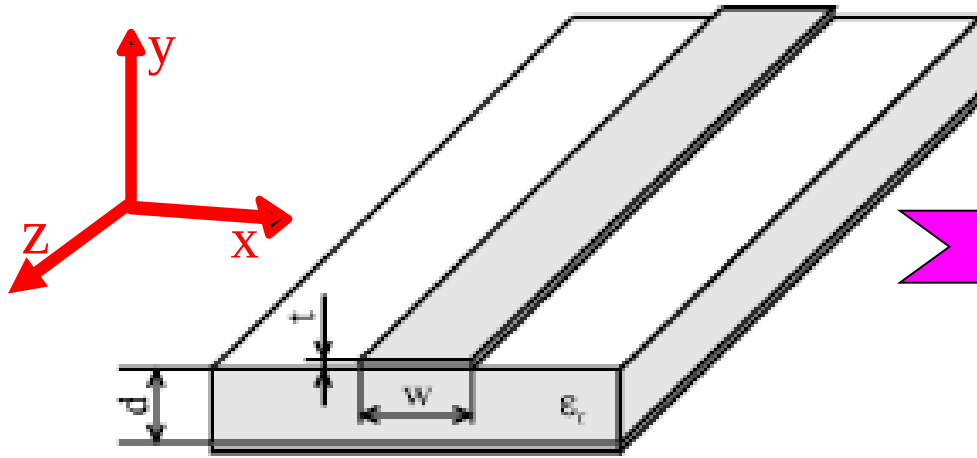
- Parallel plate waveguide
- Rectangular waveguide
- Circular waveguide

Waveguides

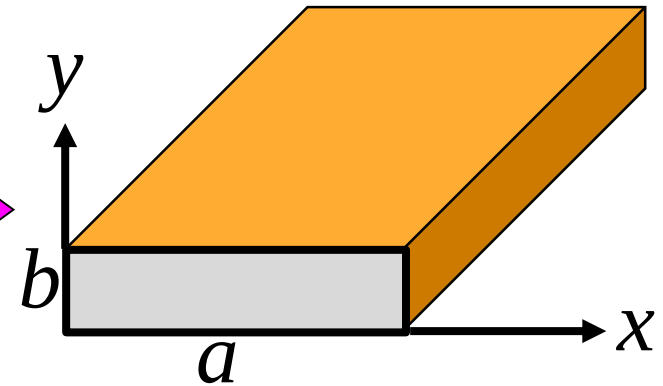
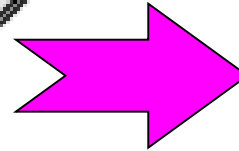
- Optical fiber (dielectric waveguide)

Note: In microwave engineering, the term “waveguide” is often used to mean rectangular or circular waveguide (i.e., a hollow pipe of metal).

DRAWBACK OF TRANSMISSION LINES TECHNOLOGIES



MICROSTRIP LINES

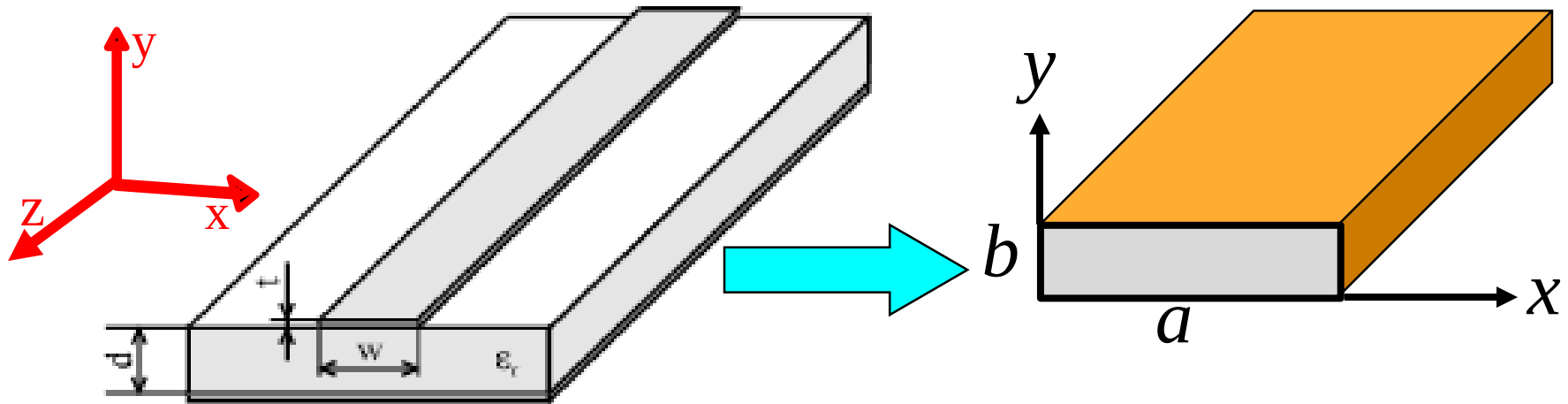


WAVEGUIDE

Disadvantages of Microstrip Lines

- ❖ **Low power-handling** capability due to small size.
- ❖ Radiation losses from the open structures like microstrip, slot and coplanar lines.
- ❖ This loss is reduced by confining the field lines more in the dielectric substrate of a high dielectric constant.

DRAWBACK OF TRANSMISSION LINES TECHNOLOGIES

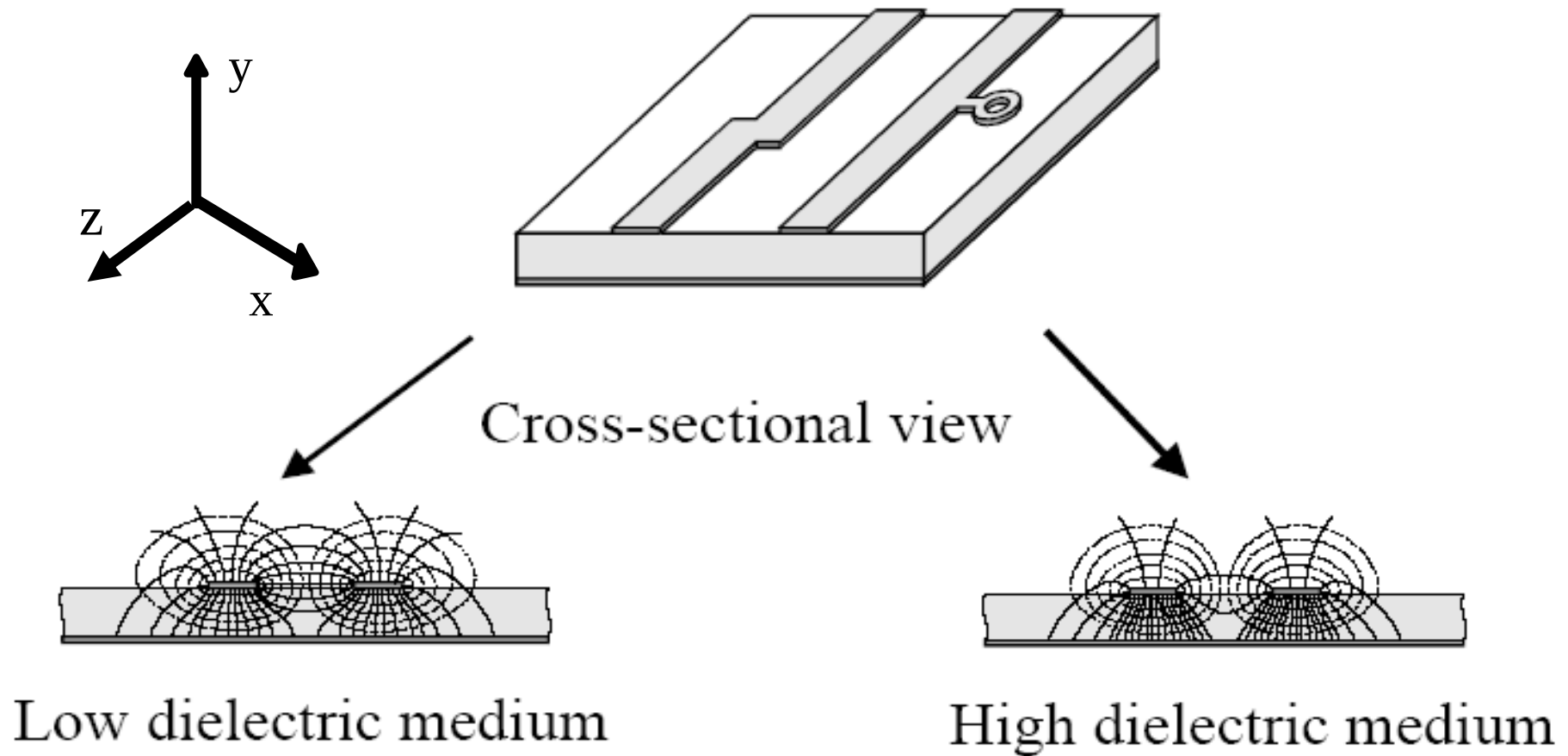


MICROSTRIP LINES

WAVEGUIDE

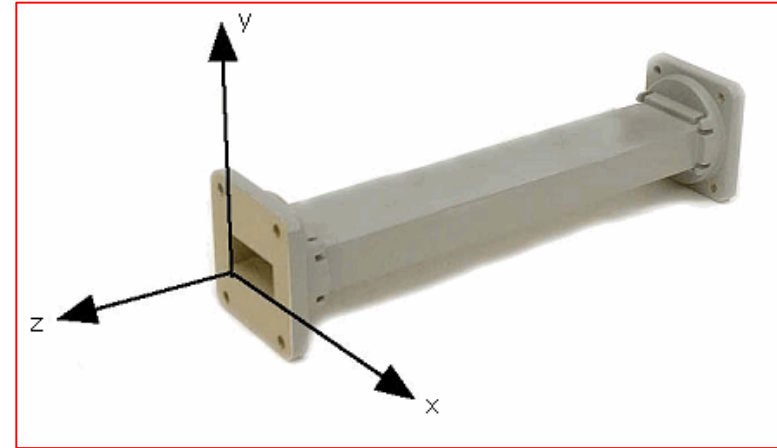
- ❖ Substrate thickness d
- ❖ Width of metal conductor strip is w ,
- ❖ Condition of wave propagation only along z -direction of Microstrip line is
 - $w \ll \lambda$ (no propagation along x -direction)
 - $d \ll \lambda$ (no propagation along y -direction)

EXAMPLE OF TRANSMISSION LINE: MICROSTIP LINE



RECTANGULAR WAVEGUIDE

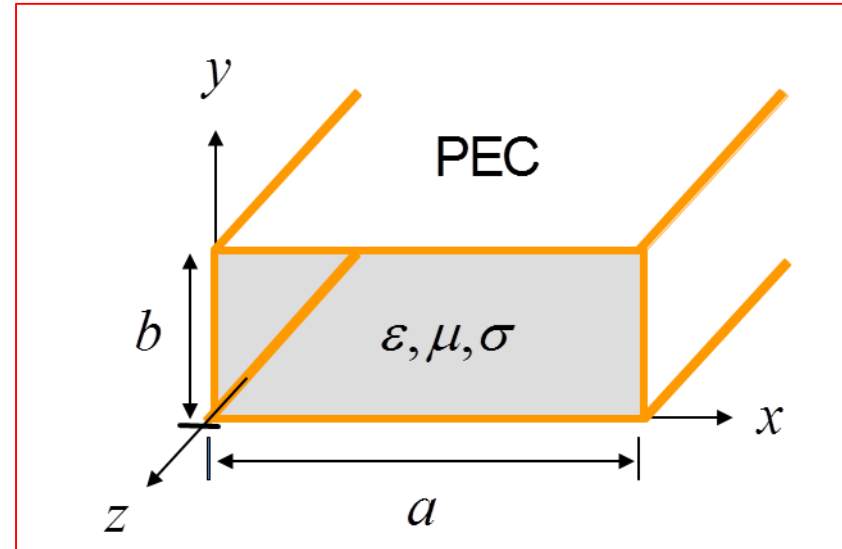
- One of the earliest waveguides.
- Still common for high power and low-loss microwave / millimeter-wave applications.



- It is essentially an electromagnetic pipe with a rectangular cross-section.

Single conductor $\Rightarrow$ No TEM mode

For convenience:



- $a \geq b$ (the long dimension lies along x).

USES OF RECTANGULAR WAVEGUIDE

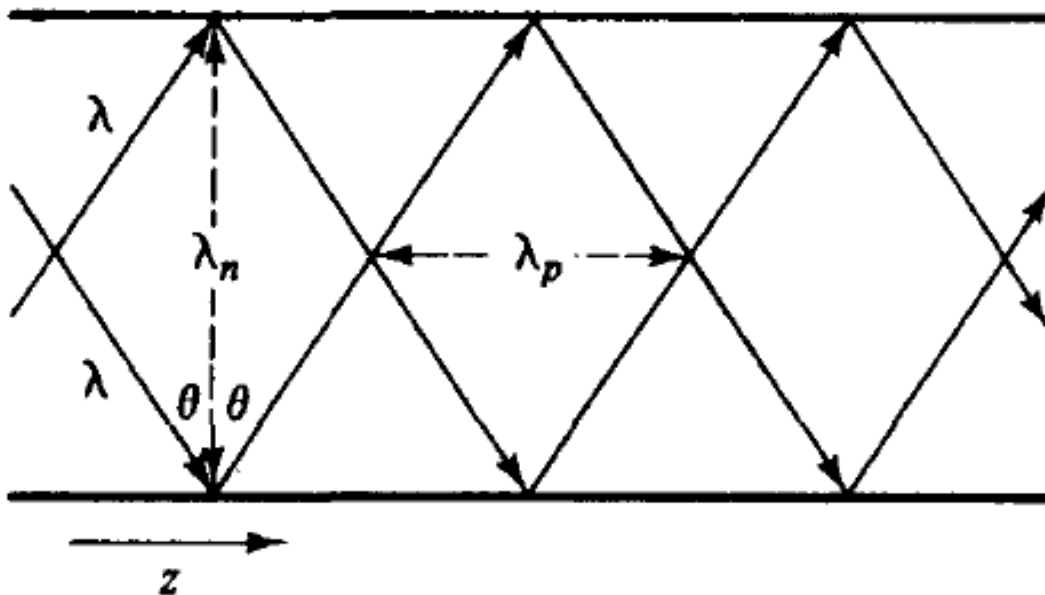
- To reduce attenuation loss
 - High frequencies
 - High power
- Can operate only above certain frequencies
 - Acts as a High-pass filter
- Normally circular or rectangular
 - We will assume lossless rectangular

THE PROCESS OF SOLVING THE WAVEGUIDE PROBLEMS MAY INVOLVE THREE STEPS

- The desired wave equations are written in the form of either rectangular or cylindrical coordinate systems suitable to the problem at hand.
- The boundary conditions are then applied to the wave equations set up in step 1.
- The resultant equations usually are in the form of partial differential equations in either time or frequency domain. They can be solved by using the proper method.

RECTANGULAR WAVEGUIDE (CONTD.)

- When the waves travel longitudinally down the guide, the plane waves are reflected from wall to wall.
- This process results in a component of either electric or magnetic field in the direction of propagation of the resultant wave.



Plane wave
reflected in a z
directed
waveguide

RECTANGULAR WAVEGUIDE (CONTD.)

- It is clear that when the wavelength λ is in the direction of propagation of the incident wave, there will be one component λ_n in the direction normal to the reflecting plane and another λ_p parallel to the plane.
- These components are where $\theta =$ angle of incidence $\lambda =$ wavelength of the impressed signal in unbounded medium

$$\lambda_n = \frac{\lambda}{\cos \theta}$$

and

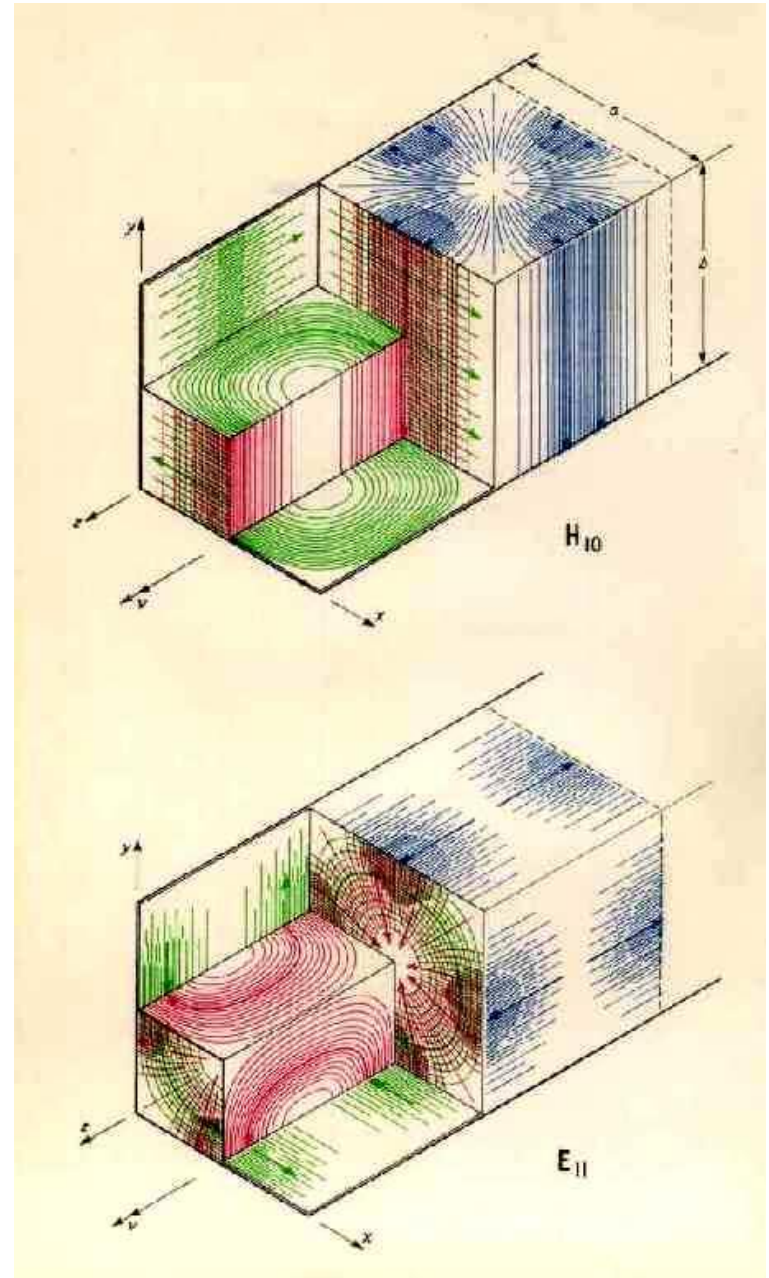
$$\lambda_p = \frac{\lambda}{\sin \theta}$$

RECTANGULAR WAVEGUIDE (CONTD.)

- ❖ Need to find the fields components of the *em* wave inside the waveguide

$$-E_z \quad H_z \quad E_x \quad H_x \quad E_y \quad H_y$$

- ❖ We'll find that *waveguides* don't support TEM waves



RECTANGULAR WAVEGUIDE (CONTD.)

Using phasors & assuming waveguide filled with

- ❖ Lossless dielectric material and
- ❖ Walls of perfect conductor,
- ❖ The wave inside should obey

$$\nabla^2 E + k^2 E = 0$$

$$\nabla^2 H + k^2 H = 0$$

where $k^2 = \omega^2 \mu \epsilon_c$

- ❖ Then applying on the **z-component**

$$\nabla^2 E_z + k^2 E_z = 0$$

RECTANGULAR WAVEGUIDE (CONTD.)

❖ Fields inside the waveguide

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \frac{\partial^2 E_z}{\partial z^2} + k^2 E_z = 0$$

Derivation can
be ask in
EXAM

Solving by method of Separation of Variables :

$$E_z(x, y, z) = X(x)Y(y)Z(z)$$

from where we obtain :

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = -k^2$$

$$h^2 = \gamma^2 + k^2 = k_x^2 + k_y^2$$

RECTANGULAR WAVEGUIDE (CONTD.)

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = -k^2$$
$$-k_x^2 - k_y^2 + \gamma^2 = -k^2$$

$$X'' + k_x^2 X = 0$$

$$Y'' + k_y^2 Y = 0$$

$$Z'' - \gamma^2 Z = 0$$

$$X(x) = c_1 \cos k_x x + c_2 \sin k_x x$$

$$Y(y) = c_3 \cos k_y y + c_4 \sin k_y y$$

$$Z(z) = c_5 e^{\gamma z} + c_6 e^{-\gamma z}$$

RECTANGULAR WAVEGUIDE (CONTD.)

❖ Substituting

$$E_z(x, y, z) = X(x)Y(y)Z(z) \begin{cases} X(x) = c_1 \cos k_x x + c_2 \sin k_x x \\ Y(y) = c_3 \cos k_y y + c_4 \sin k_y y \\ Z(z) = c_5 e^{\gamma z} + c_6 e^{-\gamma z} \end{cases}$$

$$h_z(x, y) = \overbrace{(A_1 \cos k_x x + A_2 \sin k_x x)}^{X(x)} \overbrace{(A_3 \cos k_y y + A_4 \sin k_y y)}^{Y(y)}$$

$$E_z = (A_1 \cos k_x x + A_2 \sin k_x x)(A_3 \cos k_y y + A_4 \sin k_y y)(A_5 e^{\gamma z} + A_6 e^{-\gamma z})$$

If only looking at the wave traveling in +z-direction:

$$E_z = (A_1 \cos k_x x + A_2 \sin k_x x)(A_3 \cos k_y y + A_4 \sin k_y y)e^{-\gamma z}$$

Similarly for the magnetic field,

$$H_z = (B_1 \cos k_x x + B_2 \sin k_x x)(B_3 \cos k_y y + B_4 \sin k_y y)e^{-\gamma z}$$

RECTANGULAR WAVEGUIDE (CONTD.)

IF we know E_z and H_z , we can find all the other fields from Faraday and Ampere Laws

$$E_x = -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

$$E_y = -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial y} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x}$$

$$H_x = \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial y} - \frac{\gamma}{h^2} \frac{\partial H_z}{\partial x}$$

$$H_y = -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x} - \frac{\gamma}{h^2} \frac{\partial H_z}{\partial y}$$

where

$$h^2 = \gamma^2 + k^2 = k_x^2 + k_y^2$$

RECTANGULAR WAVEGUIDE (CONTD.)

MODES OF PROPAGATION

TEM ($E_z = H_z = 0$)

can't propagate through the waveguide.

TE ($E_z = 0$) transverse electric, H_z exists

In TE mode, the electric lines of flux are perpendicular to the axis of the waveguide

TM ($H_z = 0$) transverse magnetic, E_z exists

In TM mode, the magnetic lines of flux are perpendicular to the axis of the waveguide.

HE hybrid modes in which all components exists

TE_z MODES (CONT.)

Reason for non-physical solution

Start with the vector wave equation:

$$\nabla \times (\nabla \times \underline{H}) - k^2 \underline{H} = \underline{0}$$

Vector wave equation:
from Maxwell's equations.



Take divergence of both sides.

$$\cancel{\nabla \cdot} (\nabla \times (\nabla \times \underline{H})) - k^2 \nabla \cdot \underline{H} = 0$$

The divergence of a curl is zero.



$$\nabla \cdot \underline{H} = 0$$

Magnetic Gauss law

TE_z MODES (CONT.)

Reason for non-physical solution

Revisit how we obtained the vector Helmholtz equation:

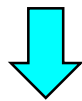
$$\nabla \times (\nabla \times \underline{H}) - k^2 \underline{H} = \underline{0} \quad \text{Vector wave equation: from Maxwell's equations.}$$



$$\underbrace{\nabla (\nabla \cdot \underline{H}) - \nabla^2 \underline{H}}_{\nabla \times \nabla \times \underline{H}} - k^2 \underline{H} = \underline{0} \quad \text{From definition of vector Laplacian}$$

Now use:

$$\nabla \cdot \underline{H} = 0 \quad \text{Magnetic Gauss law}$$



$$\nabla^2 \underline{H} + k^2 \underline{H} = \underline{0}$$

A needed assumption!

Vector Helmholtz equation
(what we have solved)

TE_z MODES (CONT.)

Reason for non-physical solution

Vector wave equation $\Rightarrow$ magnetic Gauss law

Vector Helmholtz equation $\nRightarrow$ magnetic Gauss law

The vector Helmholtz equation does not guarantee that the magnetic Gauss law is satisfied. In the mathematical derivation, we need to assume the magnetic Gauss law in order to arrive at the vector Helmholtz equation.

All of the modes that we get by solving the Helmholtz equation should be checked to make sure that they do satisfy the magnetic Gauss law.

Note: The TE₀₀ mode is the only one that violates the magnetic Gauss law.

TM MODE

$$E_z = (A_1 \cos k_x x + A_2 \sin k_x x)(A_3 \cos k_y y + A_4 \sin k_y y)e^{-\gamma z}$$

Boundary conditions:
$$\begin{cases} E_z = 0 & \text{at } y = 0, b \\ E_z = 0 & \text{at } x = 0, a \end{cases}$$

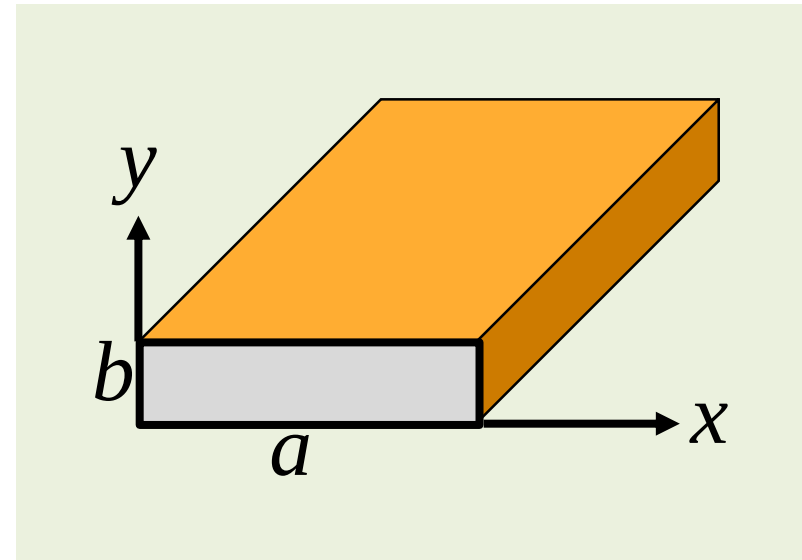
From these, we conclude:

❖ $X(x)$ is in the form of $\sin k_x x$,
where $k_x = m\pi/a$, $m=1,2,3,\dots$

❖ $Y(y)$ is in the form of $\sin k_y y$,
where $k_y = n\pi/b$, $n=1,2,3,\dots$

So the solution for $E_z(x,y,z)$ is

$$E_z = A_2 A_4 (\sin k_x x)(\sin k_y y)e^{-\gamma z}$$



TM MODE (CONTD.)

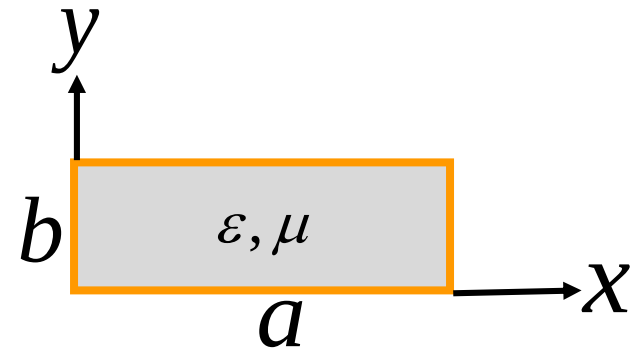
Denote

$$\frac{X''}{X} = -k_x^2 = \text{constant}$$

General solution: $X(x) = A_1 \sin(k_x x) + A_2 \cos(k_x x)$

Boundary conditions:

$$\begin{cases} X(0) = 0 & (1) \\ X(a) = 0 & (2) \end{cases}$$



(1) $\Rightarrow A_2 = 0 \Rightarrow X(x) = A_1 \sin(k_x x)$

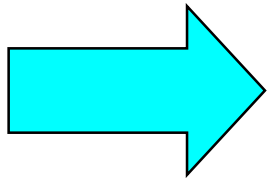
(2) $\Rightarrow \sin(k_x a) = 0$

TM MODE (CONTD.)

From the last slide: $\sin(k_x a) = 0$

This gives us the following result:

$$k_x a = m\pi, \quad m = 1, 2, \dots$$



$$k_x = \frac{m\pi}{a}$$

Hence $X(x) = A_1 \sin\left(\frac{m\pi x}{a}\right)$

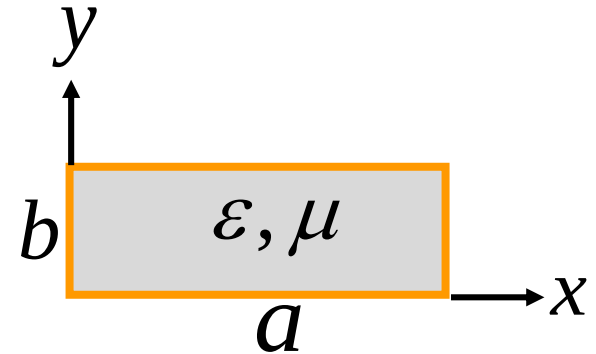
Now we turn our attention to the $Y(y)$ function.

TM MODE (CONTD.)

$$Y(y) = A_3 \sin(k_y y) + A_4 \cos(k_y y)$$

Boundary conditions:

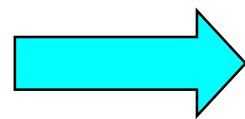
$$\begin{cases} Y(0) = 0 & (3) \\ Y(b) = 0 & (4) \end{cases}$$



$$(3) \Rightarrow A_4 = 0 \Rightarrow Y(y) = A_3 \sin(k_y y)$$

$$(4) \Rightarrow \sin(k_y b) = 0$$

Equation (4) gives us the following result: $k_y b = n\pi, \quad n = 1, 2, \dots$



$$k_y = \frac{n\pi}{b}$$

TM MODE (CONTD.)

The $Y(y)$ function is then $Y(y) = A_3 \sin\left(\frac{n\pi y}{b}\right)$

Therefore, we have

$$E_z(x, y, z=0) = X(x)Y(y) = A_1 A_3 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)$$

New notation:

$$E_z(x, y, z=0) = E_o \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)$$

The E_z field inside the waveguide thus has the following form:

$$E_z(x, y, z) = E_o \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

TM MODE

❖ Substituting

$$E_z = E_o \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) e^{-\gamma z}$$

where

$$h^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 = \gamma^2 + k^2$$

TM_{mn}

For TM_{mn}

$$E_z = E_o \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-\gamma z}$$

$$H_z = 0$$

Other components are

$$E_x = -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial x}$$

$$E_y = -\frac{\gamma}{h^2} \frac{\partial E_z}{\partial y}$$

$$H_x = \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial y}$$

$$H_y = -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x}$$

$$E_x = -\frac{\gamma}{h^2} \left(\frac{m\pi}{a}\right) E_o \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$E_y = -\frac{\gamma}{h^2} \left(\frac{n\pi}{b}\right) E_o \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$H_x = \frac{j\omega\epsilon}{h^2} \left(\frac{n\pi}{b}\right) E_o \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$H_y = -\frac{j\omega\epsilon}{h^2} \left(\frac{m\pi}{a}\right) E_o \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

PROPAGATING AND NON-PROPAGATING TM MODES

Non-propagating modes:

- TM_{00} : $H_z = 0$; $E_z = 0$; $E_x = 0$; $E_y = 0$; $H_x = 0$; $H_y = 0$
- TM_{m0} : $H_z = 0$; $E_z = 0$; $E_x = 0$; $E_y = 0$; $H_x = 0$; $H_y = 0$
- TM_{0n} : $H_z = 0$; $E_z = 0$; $E_x = 0$; $E_y = 0$; $H_x = 0$; $H_y = 0$

Propagating modes:

- TM_{mn} ; $m \geq 1$ and $n \geq 1$

TM MODES

- ❖ The m and n represent the mode of propagation and indicates the number of half cycle variations of the field in the x and y directions
- ❖ Note that for the TM mode, if n or m is zero, all fields are zero.

$$\begin{aligned}\gamma &= \sqrt{(k_x^2 + k_y^2) - k^2} \\ &= \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu \epsilon}\end{aligned}$$

❖ The cutoff frequency occurs when

$$\text{When } \omega_c^2 \mu \epsilon = \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \text{ then } \gamma = \alpha + j\beta = 0$$

$$\text{or } f_c = \frac{1}{2\pi} \frac{1}{\sqrt{\mu \epsilon}} \sqrt{\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2}$$

❖ Evanescent:

$$\text{When } \omega^2 \mu \epsilon < \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \quad \gamma = \alpha \text{ and } \beta = 0$$

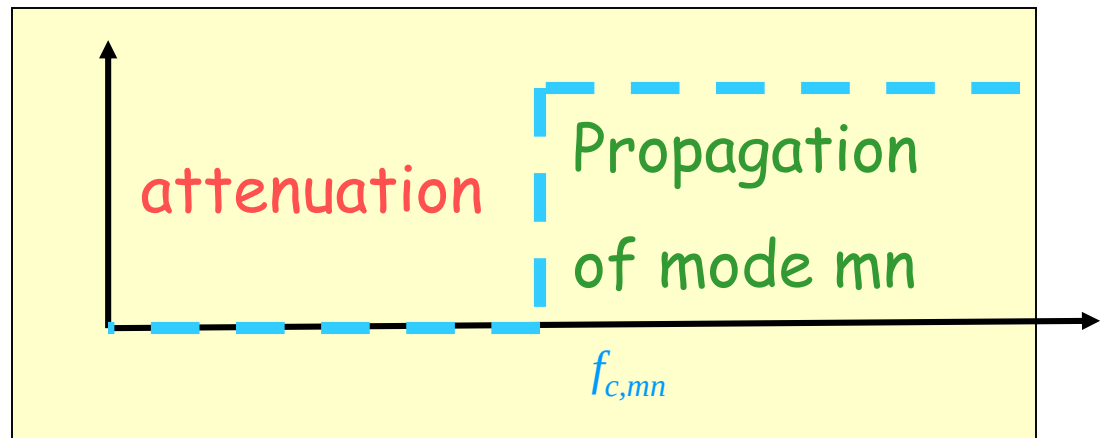
❖ Means no propagation, everything is attenuated

❖ Propagation:

$$\text{When } \omega^2 \mu \epsilon > \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \quad \gamma = j\beta \text{ and } \alpha = 0$$

❖ This is the case we are interested since is when the wave is allowed to travel through the guide

Cut-off
frequency



❖ The cutoff frequency is the frequency below which attenuation occurs and above which propagation takes place. (High Pass)

$$f_{cmn} = \frac{v}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

❖ The phase constant becomes

$$\beta = \sqrt{\omega^2 \mu \epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2} = k \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

PHASE VELOCITY AND IMPEDANCE

❖ The phase velocity is defined as

$$v = \frac{\omega}{k} \quad \lambda = \frac{2\pi}{k} = \frac{v}{f}$$

❖ And the intrinsic impedance of the mode is

$$\eta_{TM} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\beta\eta}{k} = \eta' \sqrt{1 - \left[\frac{f_c}{f} \right]^2}$$

$$\beta = k \sqrt{1 - \left(\frac{f_c}{f} \right)^2}$$

Lossless Case: Wavenumber

Above cutoff

$$f > f_c$$

$$\begin{aligned}\gamma &= k_z = \sqrt{k^2 - k_c^2} \\ &= \sqrt{\omega^2 \mu \varepsilon - \omega_c^2 \mu \varepsilon} \\ &= \omega \sqrt{\mu \varepsilon} \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} \\ &= k \sqrt{1 - \left(\frac{f_c}{f}\right)^2}\end{aligned}$$

$$\gamma = \alpha + j\beta$$

$$\beta = k \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$\alpha = 0$$

Lossless Case: Wavenumber (cont.)

Below cutoff

$$f < f_c$$

$$\gamma = \alpha + j\beta$$

$$\gamma = k_z = (k^2 - k_c^2)^{1/2}$$

$$= -j\sqrt{k_c^2 - k^2}$$

$$= -jk_c\sqrt{1 - \left(\frac{k}{k_c}\right)^2}$$

$$= -jk_c\sqrt{1 - \left(\frac{f}{f_c}\right)^2}$$

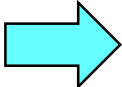
$$\beta = 0$$

$$\alpha = k_c\sqrt{1 - \left(\frac{f}{f_c}\right)^2}$$

GUIDE WAVELENGTH

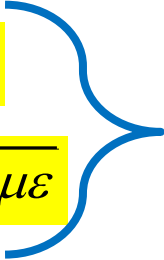
Derivation of wavelength formula (lossless waveguide):

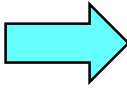
$$\lambda_g = \frac{2\pi}{\beta} = \frac{2\pi}{\sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}} = \frac{2\pi}{\sqrt{k^2 - k_c^2}}$$


$$\lambda_g = \frac{2\pi}{\sqrt{k^2 - k_c^2}} = \frac{2\pi}{k\sqrt{1 - (k_c/k)^2}} = \frac{2\pi}{\frac{2\pi}{\lambda'}\sqrt{1 - (k_c/k)^2}} = \frac{\lambda'}{\sqrt{1 - (k_c/k)^2}}$$

$$k = \omega\sqrt{\mu\epsilon} = 2\pi f\sqrt{\mu\epsilon}$$

$$k_c = \omega_c\sqrt{\mu\epsilon} = 2\pi f_c\sqrt{\mu\epsilon}$$


$$k_c/k = f_c/f$$


$$\lambda_g = \frac{\lambda'}{\sqrt{1 - (f_c/f)^2}}$$

Lossless Case: Group Velocity

Above cutoff: $f > f_c$ $\beta^2 = k^2 - k_c^2 = \omega^2 \mu \epsilon - k_c^2$

Derive both sides w.r.t β

$$2\beta \partial \beta = 2\omega \partial \omega \mu \epsilon$$

$$\beta = k \sqrt{1 - \left[\frac{f_c}{f} \right]^2}$$

so

$$v_g = \frac{d\omega}{d\beta} = \frac{1}{\mu \epsilon} \frac{\beta}{\omega} = v^2 \frac{k}{\omega} \sqrt{1 - \left[\frac{f_c}{f} \right]^2} = v^2 \frac{1}{v} \sqrt{1 - \left[\frac{f_c}{f} \right]^2} = v \sqrt{1 - \left[\frac{f_c}{f} \right]^2}$$

$$v_g = v \sqrt{1 - \left(\frac{f_c}{f} \right)^2}$$

Lossless Case: Phase Velocity

Above cutoff: $f > f_c$ $\beta^2 = k^2 - k_c^2 = \omega^2 \mu \epsilon - k_c^2$

$$v_p = \frac{\omega}{\beta} = \frac{\omega}{k \sqrt{1 - \left[\frac{f_c}{f} \right]^2}} = \frac{v}{\sqrt{1 - \left[\frac{f_c}{f} \right]^2}}$$

$$\beta = k \sqrt{1 - \left[\frac{f_c}{f} \right]^2}$$

$$v_p = \frac{v}{\sqrt{1 - \left[\frac{f_c}{f} \right]^2}}$$

$$v_p v_g = v^2$$

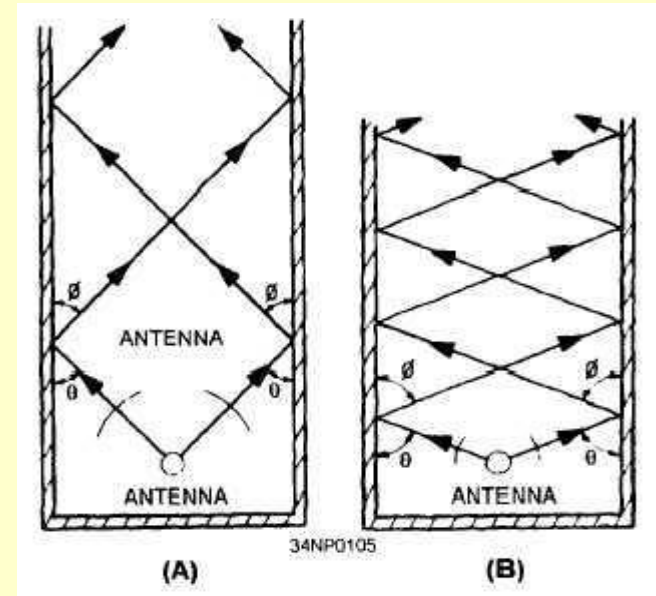
GROUP VELOCITY, U_G

❖ Is the velocity at which the energy travels.

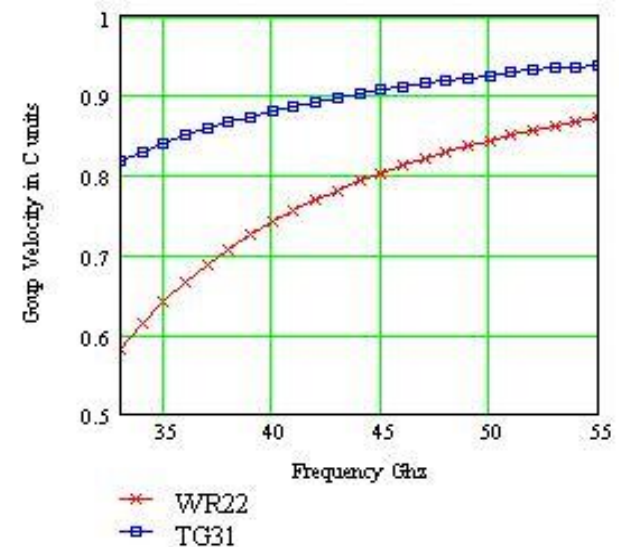
$$v_g = \frac{1}{\partial\beta / \partial\omega} = v \sqrt{1 - \left[\frac{f_c}{f} \right]^2} \quad \left[\frac{\text{rad/s}}{\text{rad/m}} \right] = \left[\frac{\text{m}}{\text{s}} \right]$$

❖ It is always less than v

$$v_p v_g = v^2$$



$$E_y = -\frac{j\omega\mu}{h^2} \left(\frac{\pi}{a} \right) H_o \sin\left(\frac{m\pi x}{a} \right) e^{-\gamma z}$$

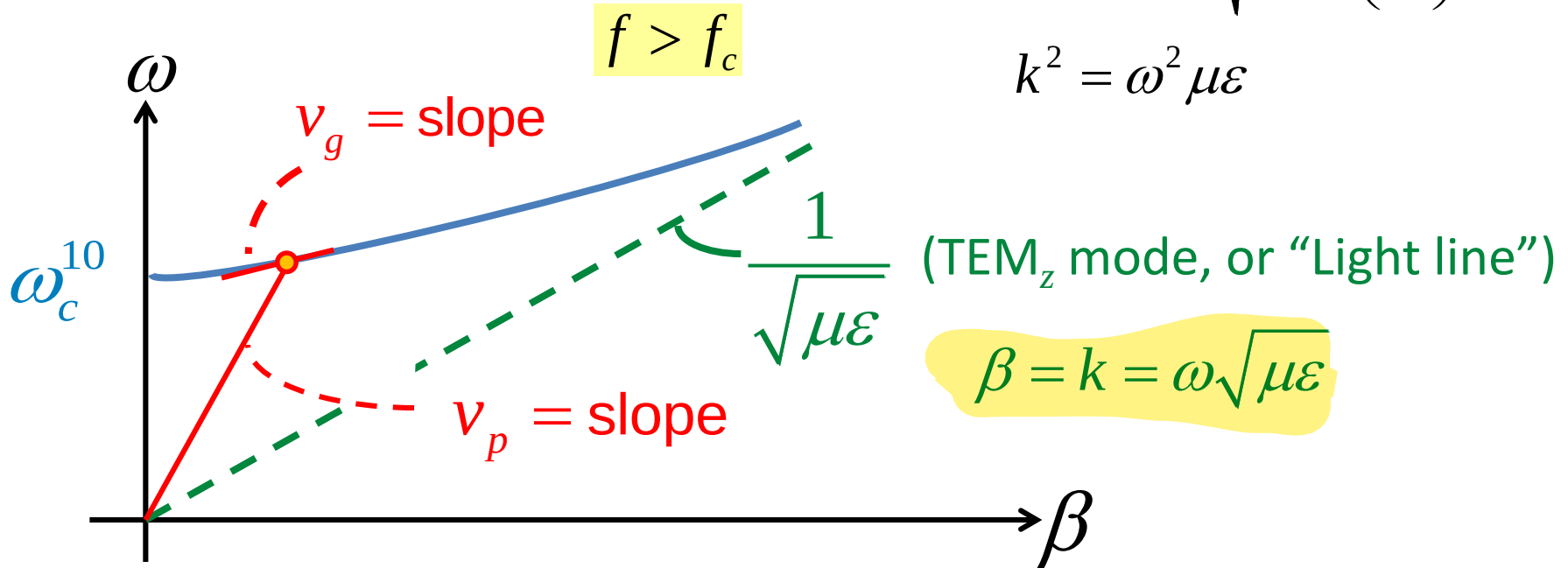


Dispersion Diagram for TE₁₀ Mode

Lossless case ($\epsilon_c = \epsilon = \epsilon'$)

$$k_z = \beta = \sqrt{k^2 - \left(\frac{\pi}{a}\right)^2}$$

$$k^2 = \omega^2 \mu \epsilon$$

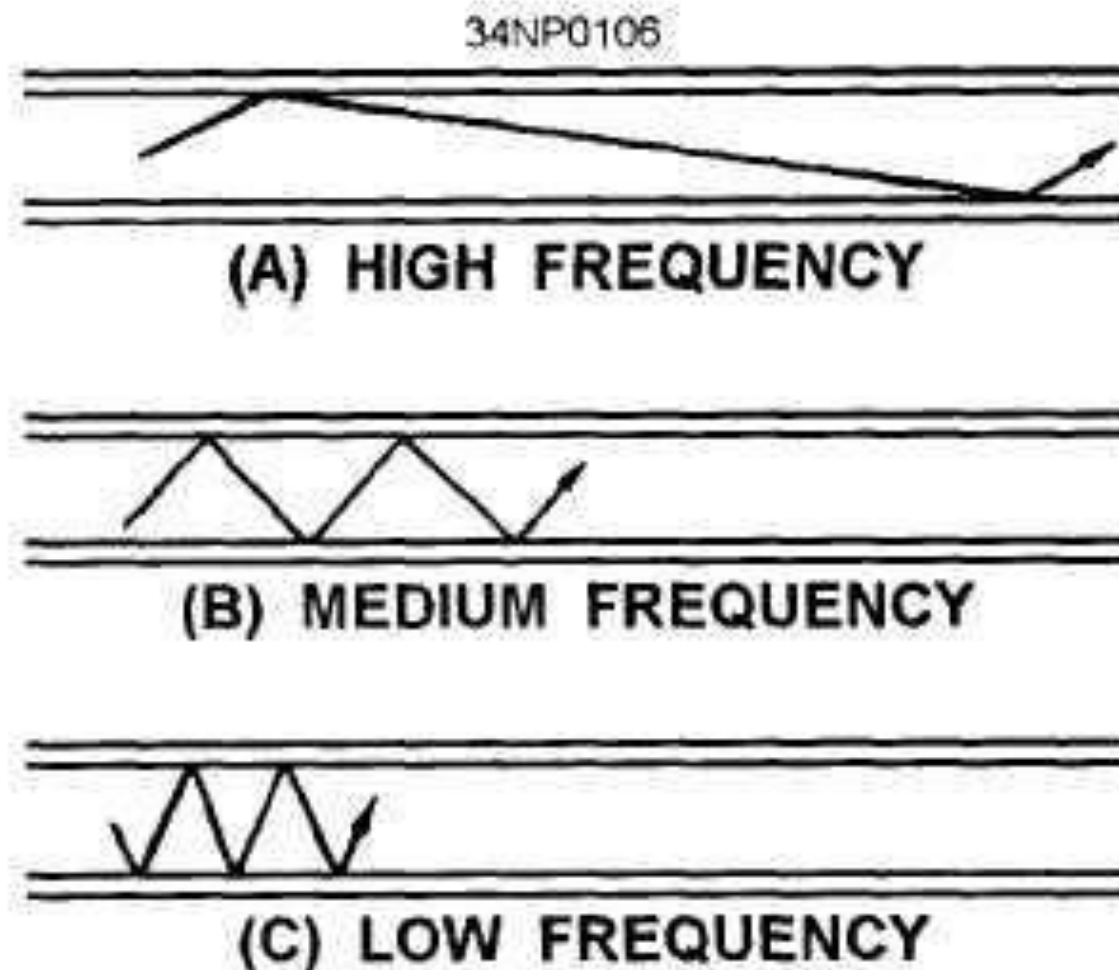


Phase velocity: $v_p = \frac{\omega}{\beta}$ Group velocity: $v_g = \frac{d\omega}{d\beta}$

Velocities are slopes on the dispersion plot.

GROUP VELOCITY

- ❖ As frequency is increased, the group velocity increases.



SUMMARY OF TM MODES

Wave in the dielectric medium	Inside the waveguide
$k = \omega / v = \omega \sqrt{\mu \epsilon}$	$\beta = k \sqrt{1 - \left[\frac{f_c}{f} \right]^2}$
$\eta' = \sqrt{\mu / \epsilon}$	$\eta_{TM} = \eta' \sqrt{1 - \left[\frac{f_c}{f} \right]^2}$
$v = \omega / k = f \lambda' = 1 / \sqrt{\mu \epsilon}$	$v_p = \frac{\omega}{k \sqrt{1 - \left[\frac{f_c}{f} \right]^2}} = \omega / \beta$
$\lambda' = v / f = \frac{\lambda_0}{\sqrt{\epsilon_r}}$	$\lambda_g = \frac{\lambda'}{\sqrt{1 - \left[\frac{f_c}{f} \right]^2}}$

TE MODE

$$H_z = (B_1 \cos k_x x + B_2 \sin k_x x)(B_3 \cos k_y y + B_4 \sin k_y y)e^{-\gamma z}$$

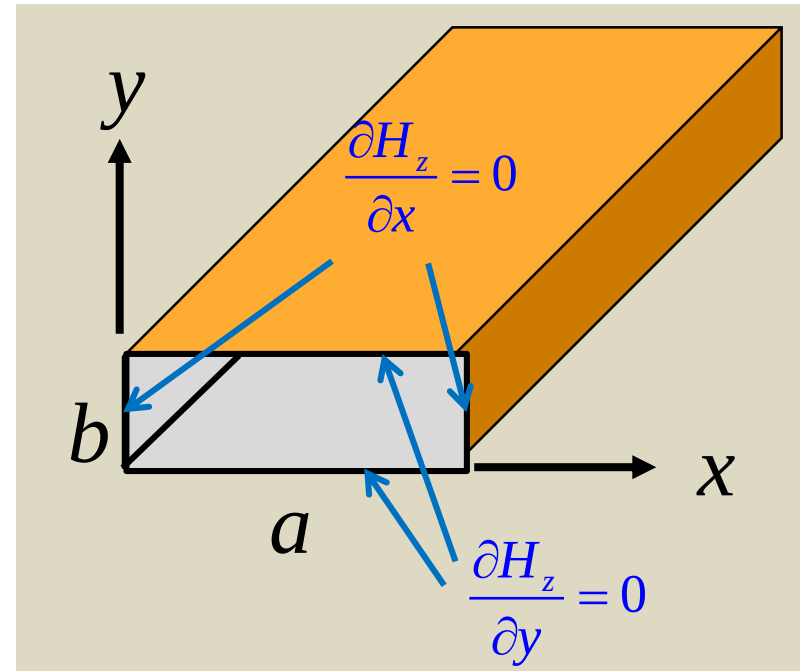
Boundary conditions:

$$\begin{aligned} E_x &= -\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y} = 0 \quad \text{at } y = 0, b \\ E_y &= -\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x} = 0 \quad \text{at } x = 0, a \end{aligned}$$

❖ From these, we conclude:

$X(x)$ is in the form of $\cos K_x x$,
where $k_x = m\pi/a$, $m=0,1,2,3,\dots$

❖ $Y(y)$ is in the form of $\cos k_y y$,
where $k_y = n\pi/b$, $n=0,1,2,3,\dots$
So the solution for $E_z(x,y,z)$ is



$$H_z = B_1 B_3 (\cos k_x x) (\cos k_y y) e^{-\gamma z}$$

TE MODE

❖ Substituting

$$H_z = H_o \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi}{b} y\right) e^{-\gamma z}$$

where again

$$h^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

- Note that n and m cannot be both zero because the fields will all be zero.

TE MODE

For TE_{mn}

$$H_z = H_o \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-\gamma z}$$

$$E_z = 0$$

❖ Other components are

$$E_x = -\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

$$E_y = -\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x}$$

$$H_x = -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial x}$$

$$H_y = -\frac{\gamma}{h^2} \frac{\partial H_z}{\partial y}$$

$$E_x = \frac{j\omega\mu}{h^2} \left(\frac{n\pi}{b}\right) H_o \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$E_y = -\frac{j\omega\mu}{h^2} \left(\frac{m\pi}{a}\right) H_o \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$H_x = \frac{j\beta}{h^2} \left(\frac{m\pi}{a}\right) H_o \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$H_y = \frac{j\beta}{h^2} \left(\frac{n\pi}{b}\right) H_o \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

PROPAGATING AND NON-PROPAGATING TM MODES

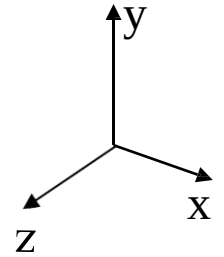
Non-propagating modes:

❖ TE_{00} : $E_z = 0$; $H_z \neq 0$; $E_x = 0$; $E_y = 0$; $H_x = 0$;
 $H_y = 0$

Propagating modes:

- ❖ TE_{0n} ; $n \geq 1$
- ❖ TE_{m0} ; $m \geq 1$
- ❖ TE_{mn} ; $m \geq 1$ and $n \geq 1$

PROPAGATING TE MODES



	TE_{10}	TE_{20}	TE_{01}
E Field			
H Field			

For Cutoff

attenuation

Propagation
of mode mn

$f_{c,mn}$

- ❖ The cutoff frequency is the same expression as for the TM mode

$$f_{c,mn} = \frac{v}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

- ❖ But the lowest attainable frequencies are lowest because here n or m can be zero.

TE MODE

❖ For TE_{mn} , wave impedance

$$Z_{TE} = \frac{E_x}{H_y} = \frac{-E_y}{H_x} = \frac{k\eta'}{k\sqrt{1-\left[\frac{f_c}{f}\right]^2}} = \frac{\eta'}{\sqrt{1-\left[\frac{f_c}{f}\right]^2}}$$

$$\beta = k\sqrt{1-\left[\frac{f_c}{f}\right]^2}$$

❖ The guide wavelength (λ : the wavelength of a plane wave in the filling medium)

$$\lambda_g = \frac{2\pi}{\beta} > \frac{2\pi}{k} = \lambda$$

$$v_p = \frac{\omega}{\beta} > \frac{\omega}{k} = 1/\sqrt{\mu\epsilon}$$

SUMMARY FOR BOTH MODES

TM_{mn}

$$E_z(x, y, z) = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$$

TE_{mn}

$$H_z(x, y, z) = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$$

$$\beta = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

same formula for both modes

$$f_{cmn} = \frac{v}{2\pi} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$v = \frac{C}{\sqrt{\epsilon_r}}$$

same formula for both modes

(lossless waveguide)

TM_z

$$\begin{aligned} m &= 1, 2, \dots \\ n &= 1, 2, \dots \end{aligned}$$

TE_z

$$\begin{aligned} m &= 0, 1, 2, \dots \\ n &= 0, 1, 2, \dots \\ (m, n) &\neq (0, 0) \end{aligned}$$

WAVENUMBER

TM_z or TE_z mode: $\beta = \sqrt{k^2 - k_c^2}$ with $k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$

Lossless waveguide:

Above cutoff:

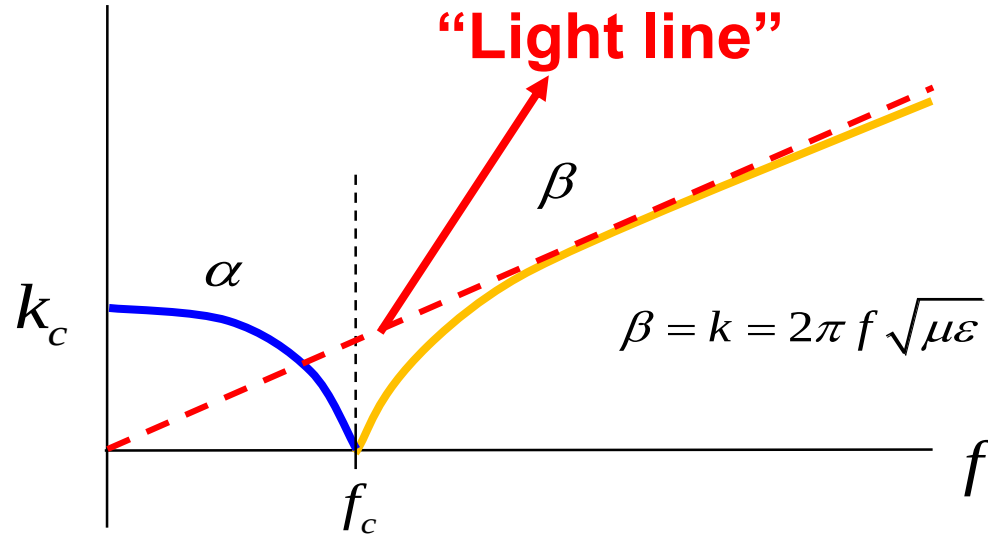
$$\beta = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Below cutoff:

$$\alpha = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - k^2}$$

WAVENUMBER PLOT

$$k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$



$$\beta = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}, \quad f > f_c$$

$$\alpha = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - k^2}, \quad f < f_c$$

$$f_c = f_{cmn} = \frac{v}{2\pi} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$v = \frac{c}{\sqrt{\epsilon_r}}$$

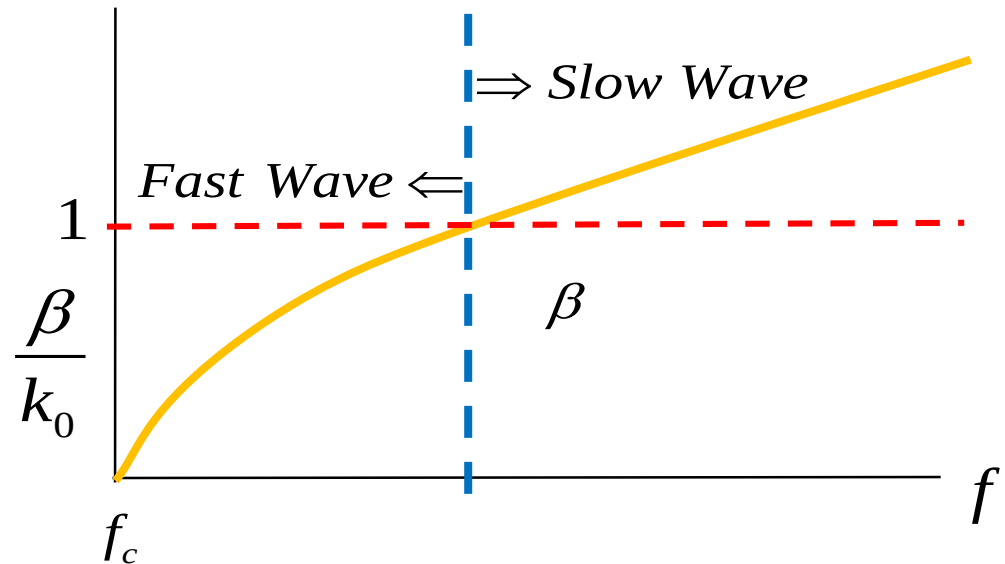
SLOW AND FAST WAVE PLOT

free space Wavenumber

$$k_0 = \frac{2\pi}{\lambda_0}$$

$$v = \frac{c}{\sqrt{\epsilon_r}}$$

v = Phase velocity of dielectric filled waveguide



⌘ β/k_0 is less than 1 so that means we have V Phase velocity higher than C or we can call this left hand side part this is the fast wave region

⌘ The right-hand side part for which β/k_0 is more than 1 or phase velocity this is less than C we call it the slow wave region.

DOMINANT MODE

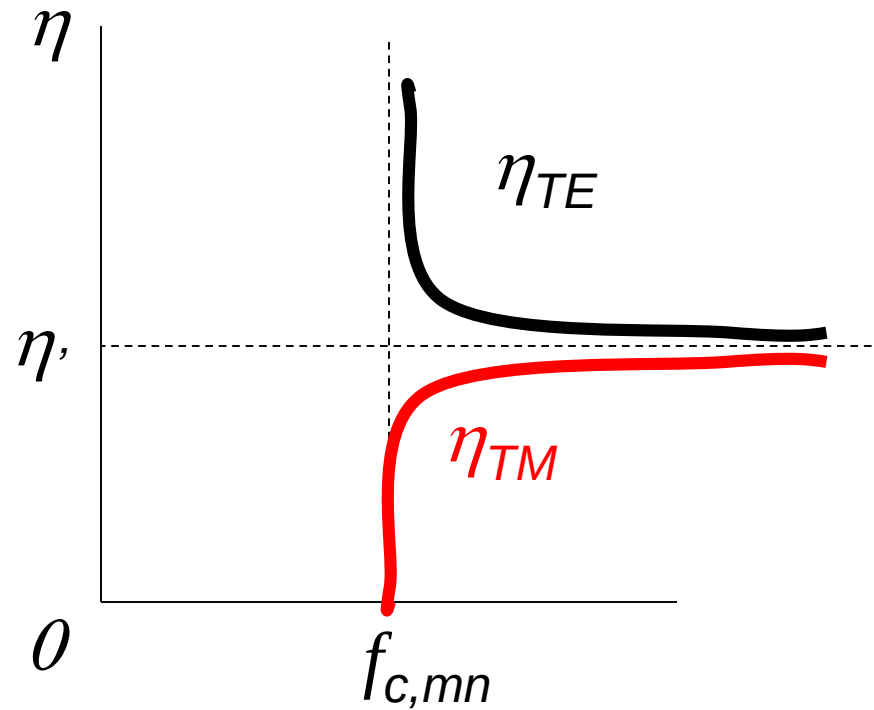
- ❖ The dominant mode is the mode with lowest cutoff frequency.
- ❖ It's always TE_{10} .
- ❖ The order of the next modes change depending on the dimensions of the guide.
- ❖ Whenever two or more modes have the same cut off frequency, they are said to be *degenerate modes*.

SUMMARY OF TE MODES

Wave in the dielectric medium	Inside the waveguide
$k = \omega / v = \omega \sqrt{\mu \epsilon}$	$\beta = k \sqrt{1 - \left[\frac{f_c}{f} \right]^2}$
$\eta' = \sqrt{\mu / \epsilon}$	$\eta_{TE} = \frac{\eta'}{\sqrt{1 - \left[\frac{f_c}{f} \right]^2}}$
$v = \omega / k = f \lambda' = 1 / \sqrt{\mu \epsilon}$	$v_p = \frac{\omega}{k \sqrt{1 - \left[\frac{f_c}{f} \right]^2}} = \omega / \beta$
$\lambda' = v / f = \frac{\lambda_0}{\sqrt{\epsilon_r}}$	$\lambda_g = \frac{\lambda'}{\sqrt{1 - \left[\frac{f_c}{f} \right]^2}}$

VARIATION OF WAVE IMPEDANCE

- ❖ Wave impedance varies with frequency and mode



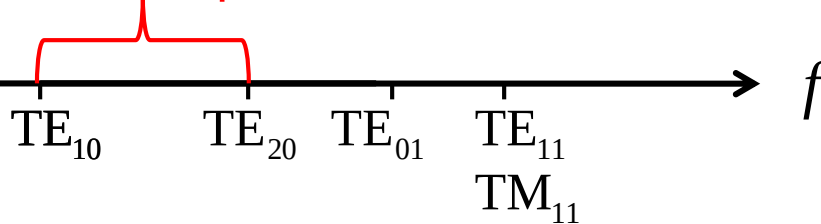
MODE CHART

Lossless case ($\epsilon_c = \epsilon = \epsilon'$)

Two cases are considered:

$$b < a / 2$$

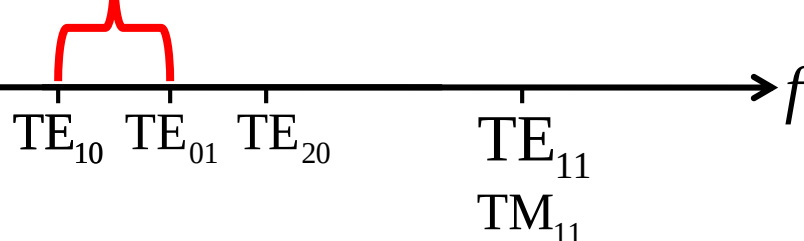
Single mode operation



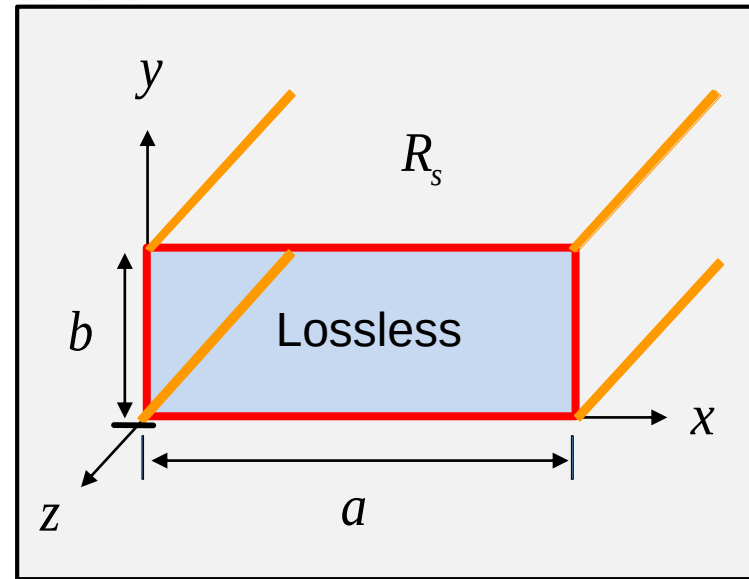
$$b > a / 2$$

Single mode operation

f_0 = center frequency



$$BW \equiv \frac{f_2 - f_1}{f_0} = \frac{(1.9 - 1.25)f_c}{f_c} = 0.65$$

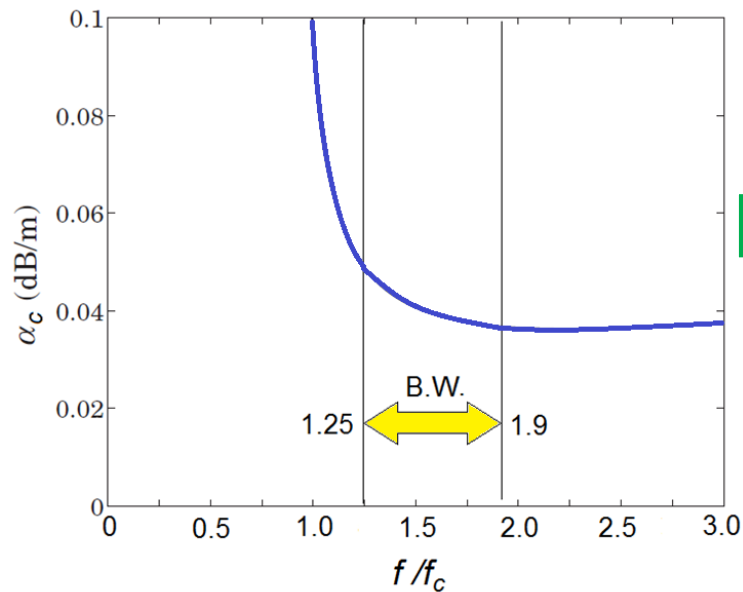


The maximum bandwidth for single-mode operation is 67%.

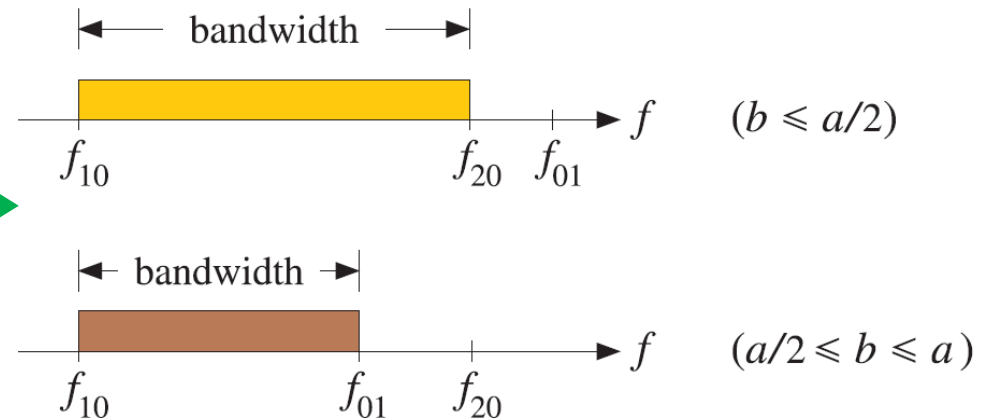
$$(b \leq a / 2)$$

$$f_c^{mn} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

RECTANGULAR WAVE GUIDES



Variation of α_c .



Operating bandwidth in rectangular waveguides.

➤ Loss increases with decreasing b .

➤ For a dielectric filled guide, attenuation (conductor + dielectric loss) is minimum approximately at $1.5f_c$ for TE_{10} mode excitation.

➤ Higher order modes offer lower loss

$$BW \equiv \frac{f_2 - f_1}{f_0} = \frac{(1.9 - 1.25)f_c}{f_c} = 0.65$$

TE_{mn} MODE

For TE_{mn}

$$\begin{aligned} h^2 \Big|_{m=1, n=0} &= \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \\ &= \left(\frac{\pi}{a} \right)^2 \end{aligned}$$

For TE₁₀

$$E_x = \frac{j\omega\mu}{h^2} \left(\frac{n\pi}{b} \right) H_o \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$E_y = -\frac{j\omega\mu}{h^2} \left(\frac{m\pi}{a} \right) H_o \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$H_x = \frac{j\beta}{h^2} \left(\frac{m\pi}{a} \right) H_o \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

$$H_y = \frac{j\beta}{h^2} \left(\frac{n\pi}{b} \right) H_o \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

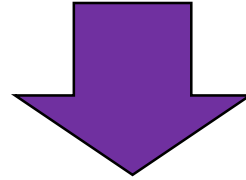
$$E_y = -\frac{j\omega\mu a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z}$$

$$H_x = \frac{j\beta a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z}$$

$$H_z = H_o \cos\left(\frac{\pi}{a} x\right) e^{-j\beta z}$$

TE_{mn} MODE

Time varying TE₁₀ fields



$$\begin{aligned}\tilde{E}_y(x, t) &= \text{Re} \left\{ E_y(x) e^{j\omega t} \right\} \\ &= \text{Re} \left\{ -\frac{j\omega\mu a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z} e^{j\omega t} \right\} \\ &= \text{Re} \left\{ -\frac{j\omega\mu a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) e^{j(\omega t - \beta z)} \right\} \\ &= \frac{\omega\mu a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) \sin(\omega t - \beta z)\end{aligned}$$

TIME VARYING FIELDS FOR TE₁₀ MODE

$$\tilde{E}_y(x, t) = \frac{\omega \mu a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) \sin(\omega t - \beta z)$$

$$\tilde{H}_x(x, t) = -\frac{\beta a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) \sin(\omega t - \beta z)$$

$$\tilde{H}_z(x, t) = H_o \cos\left(\frac{\pi x}{a}\right) \cos(\omega t - \beta z)$$

TIME VARYING FIELDS FOR TE₁₀ MODE

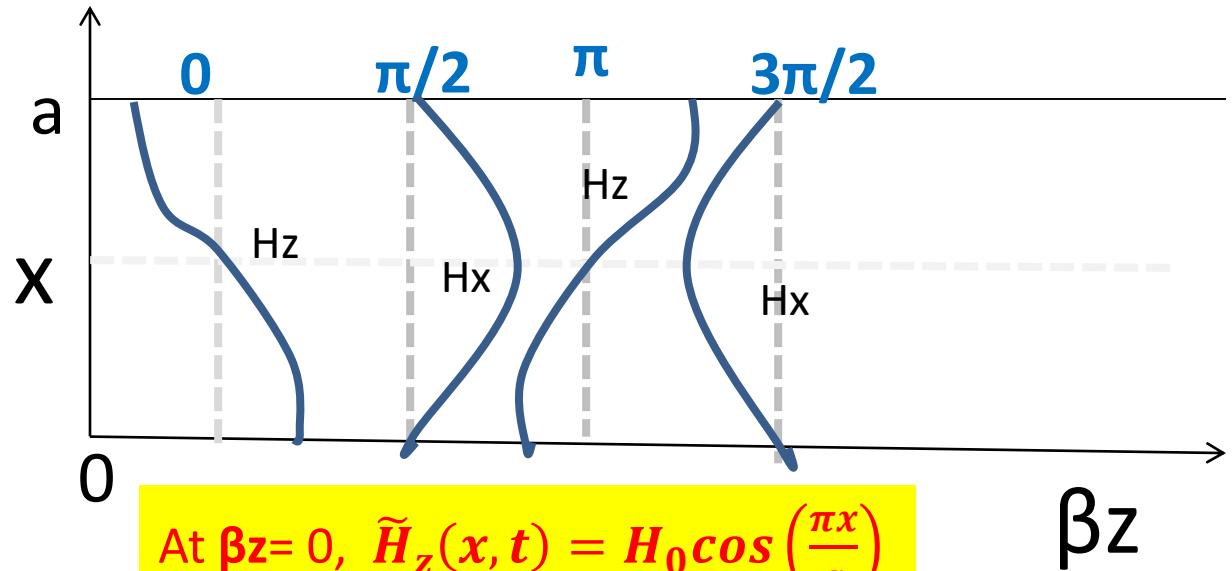
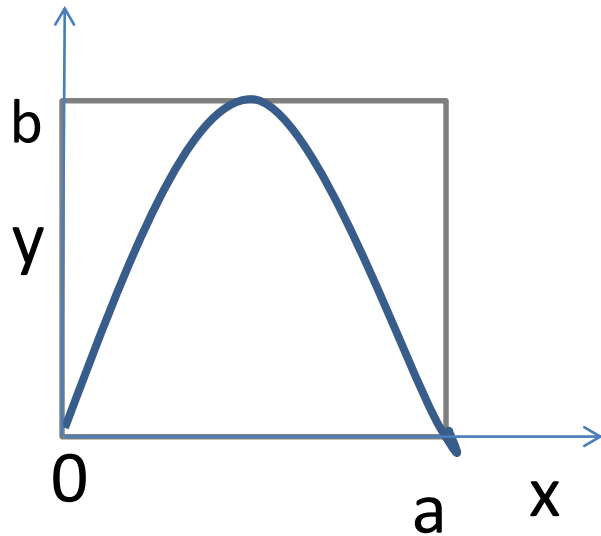
When $\omega t=0$

$$\tilde{E}_y(x, t) = -\frac{\omega \mu a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) \sin(\beta z)$$

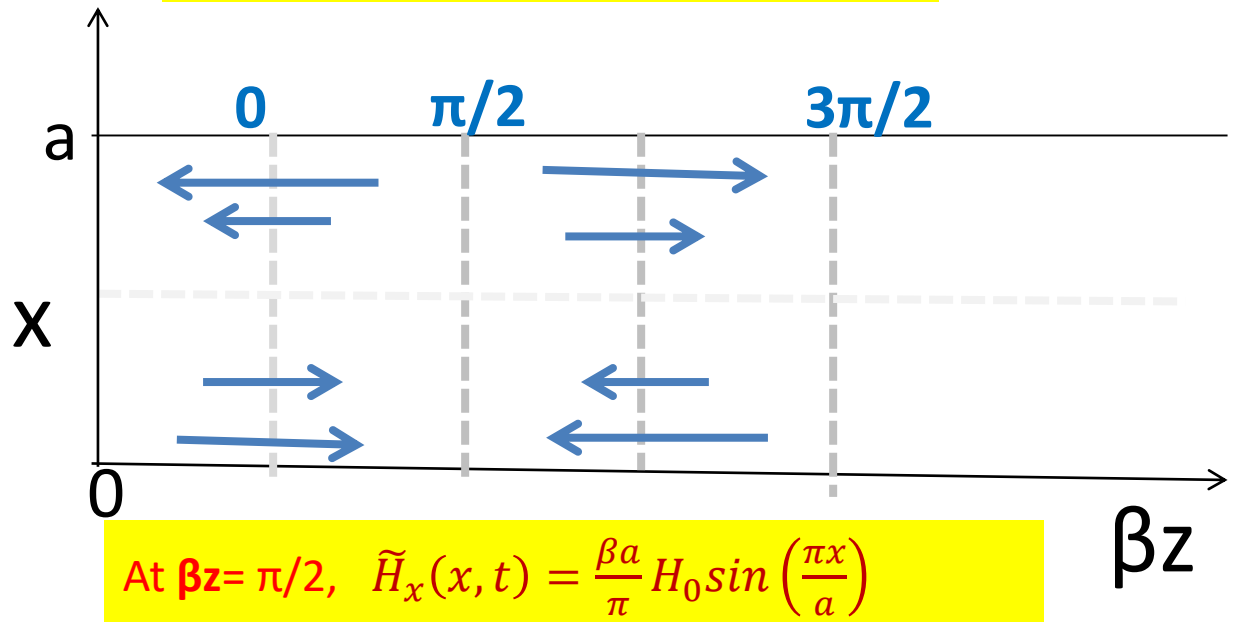
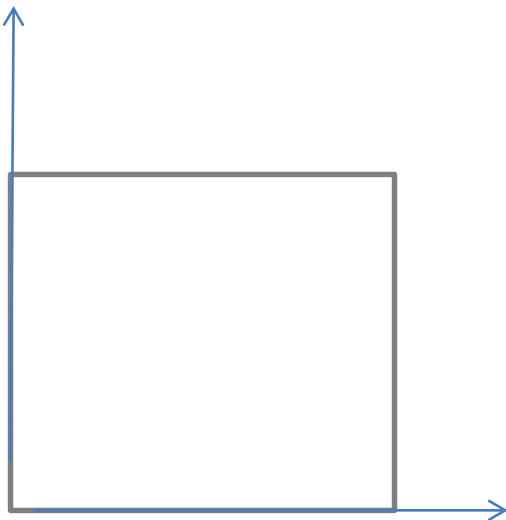
$$\tilde{H}_x(x, t) = \frac{\beta a}{\pi} H_o \sin\left(\frac{\pi x}{a}\right) \sin(\beta z)$$

$$\tilde{H}_z(x, t) = H_o \cos\left(\frac{\pi x}{a}\right) \cos(\beta z)$$

FIELDS FOR TE₁₀ MODE

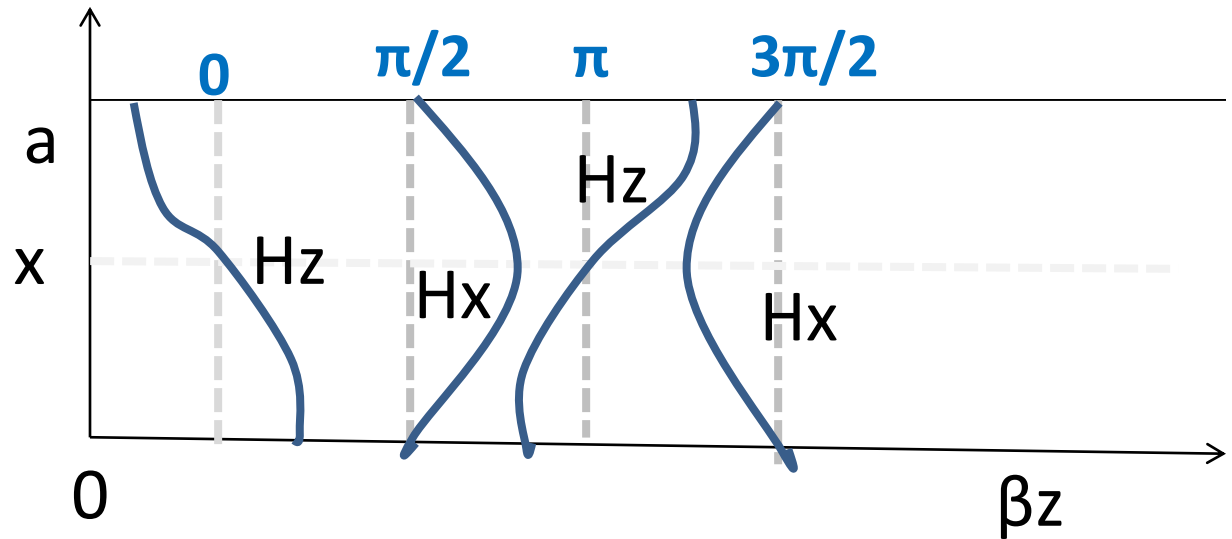
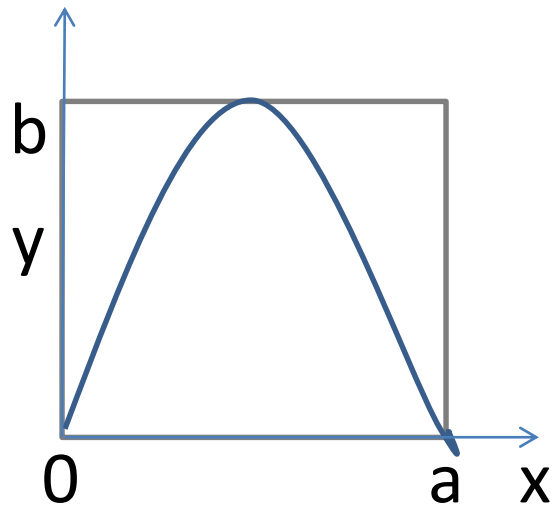


At $\beta z = 0$, $\tilde{H}_z(x, t) = H_0 \cos\left(\frac{\pi x}{a}\right)$



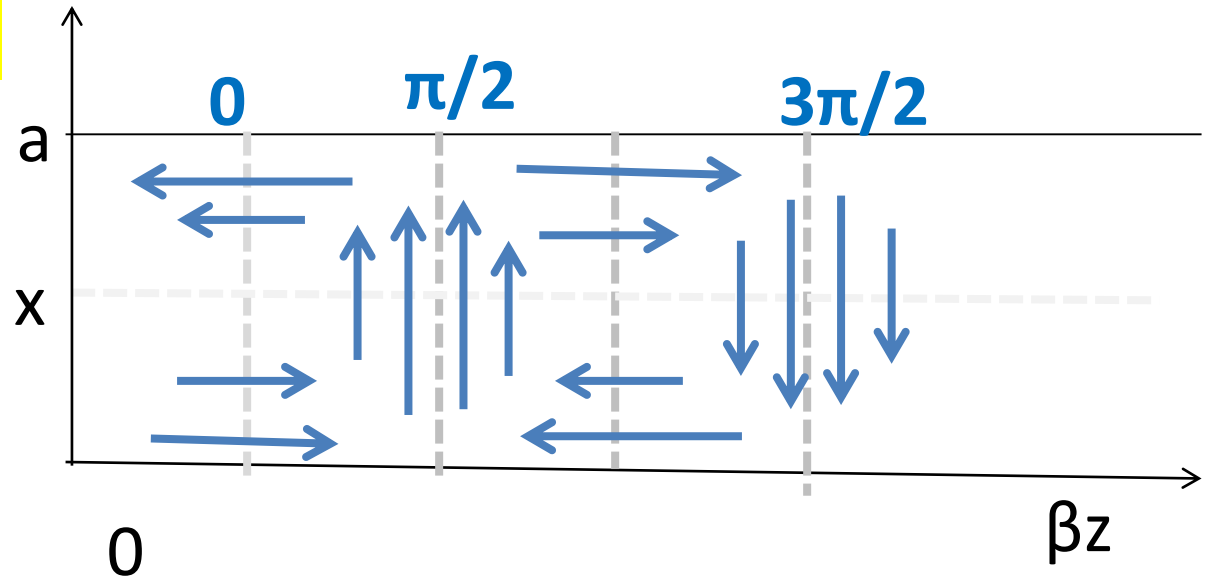
At $\beta z = \pi/2$, $\tilde{H}_x(x, t) = \frac{\beta a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right)$

FIELDS FOR TE₁₀ MODE

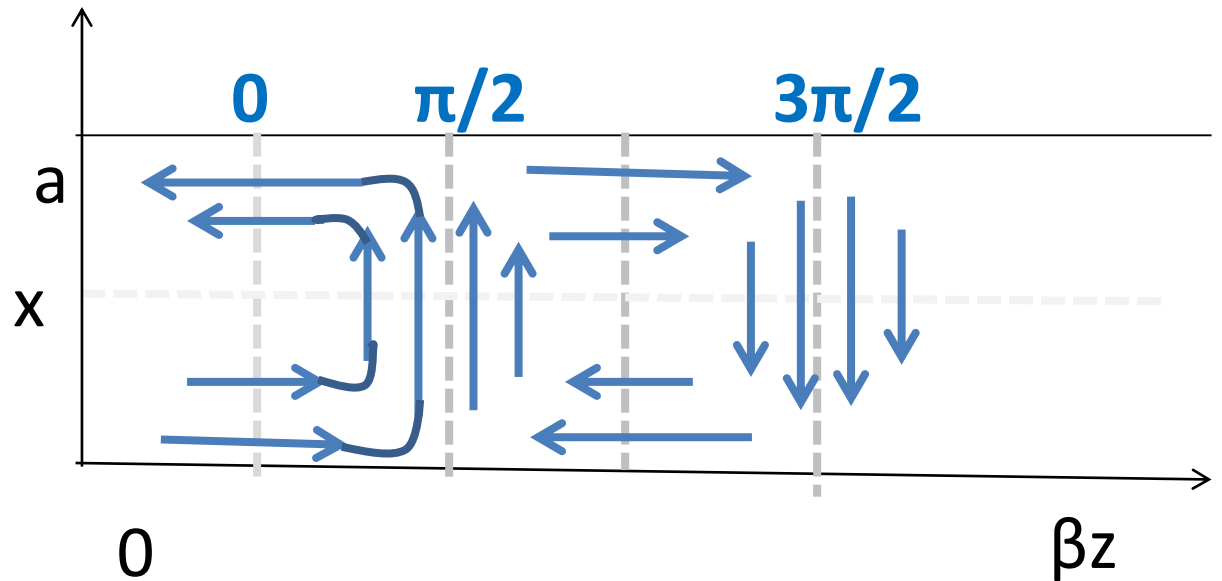
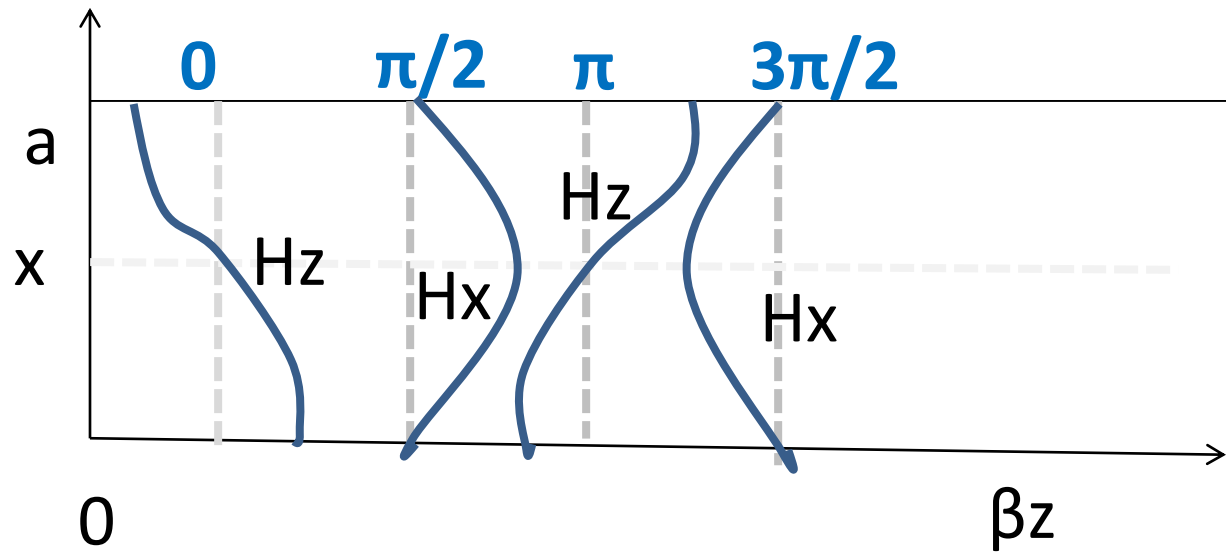
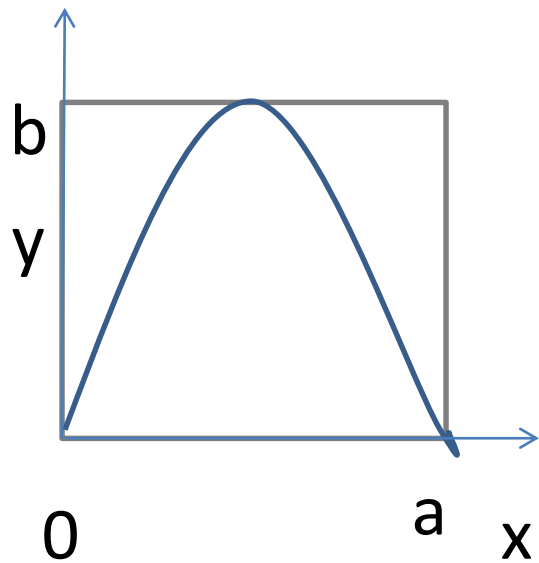


At $\beta z = \pi/2$,

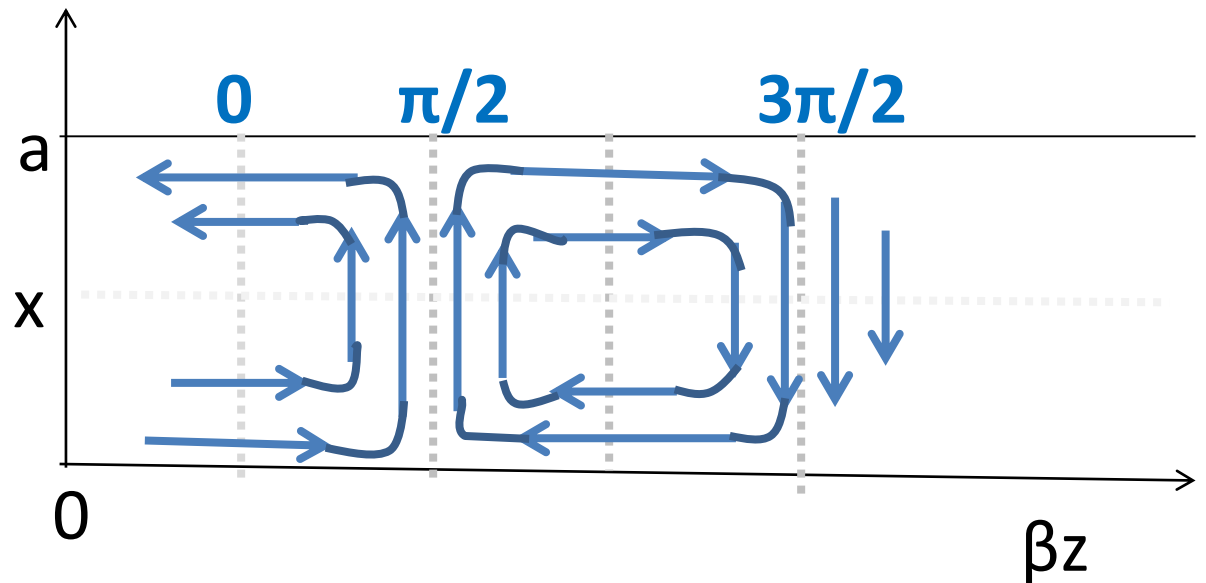
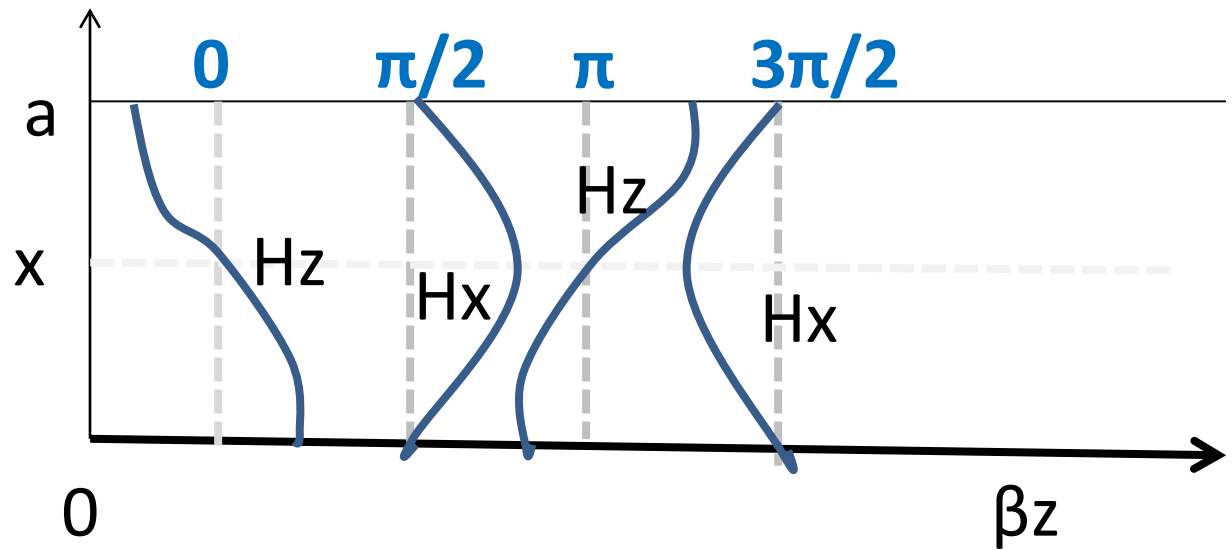
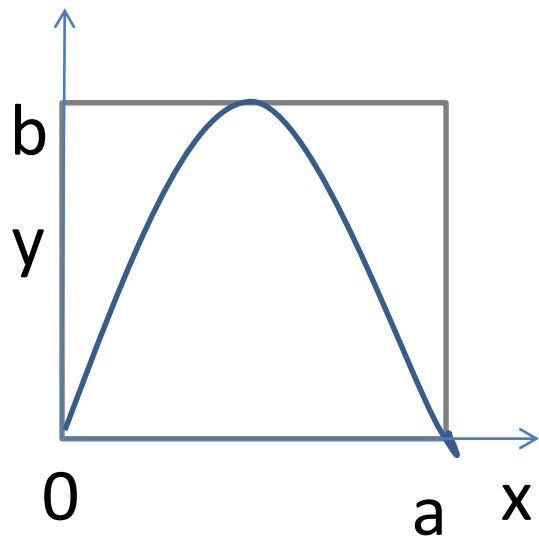
$$\tilde{H}_x(x, t) = \frac{\beta a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right)$$



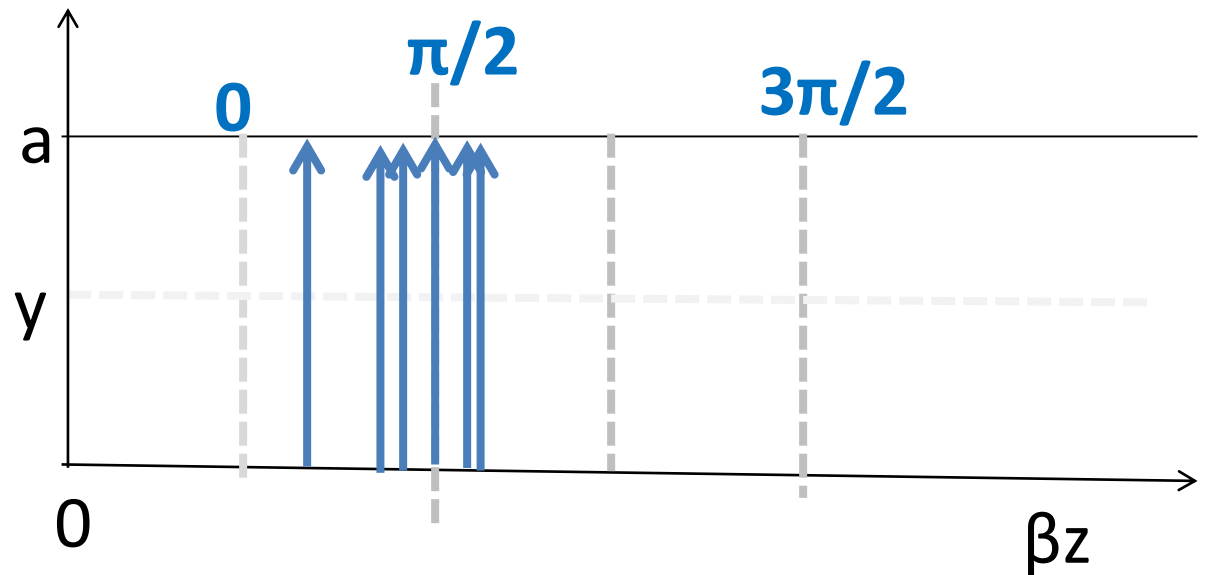
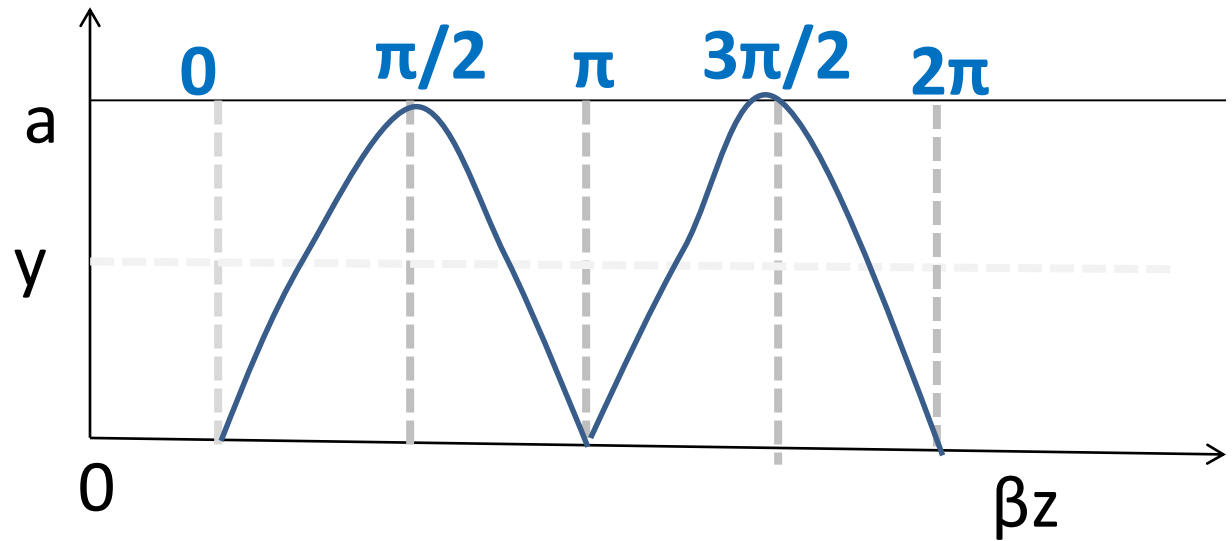
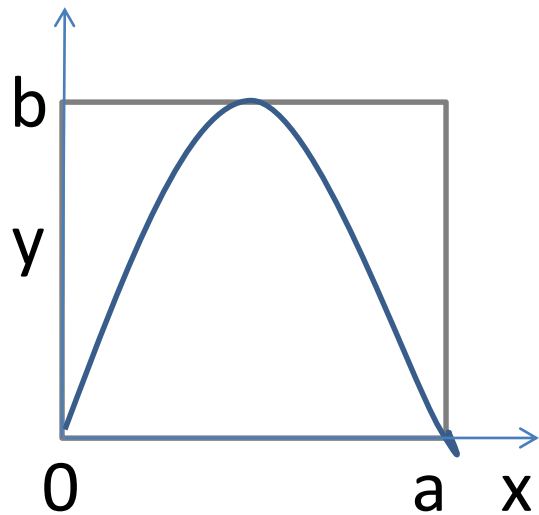
FIELDS FOR TE₁₀ MODE



FIELDS FOR TE₁₀ MODE



FIELDS FOR TE₁₀ MODE



FIELDS FOR TE₁₀ MODE

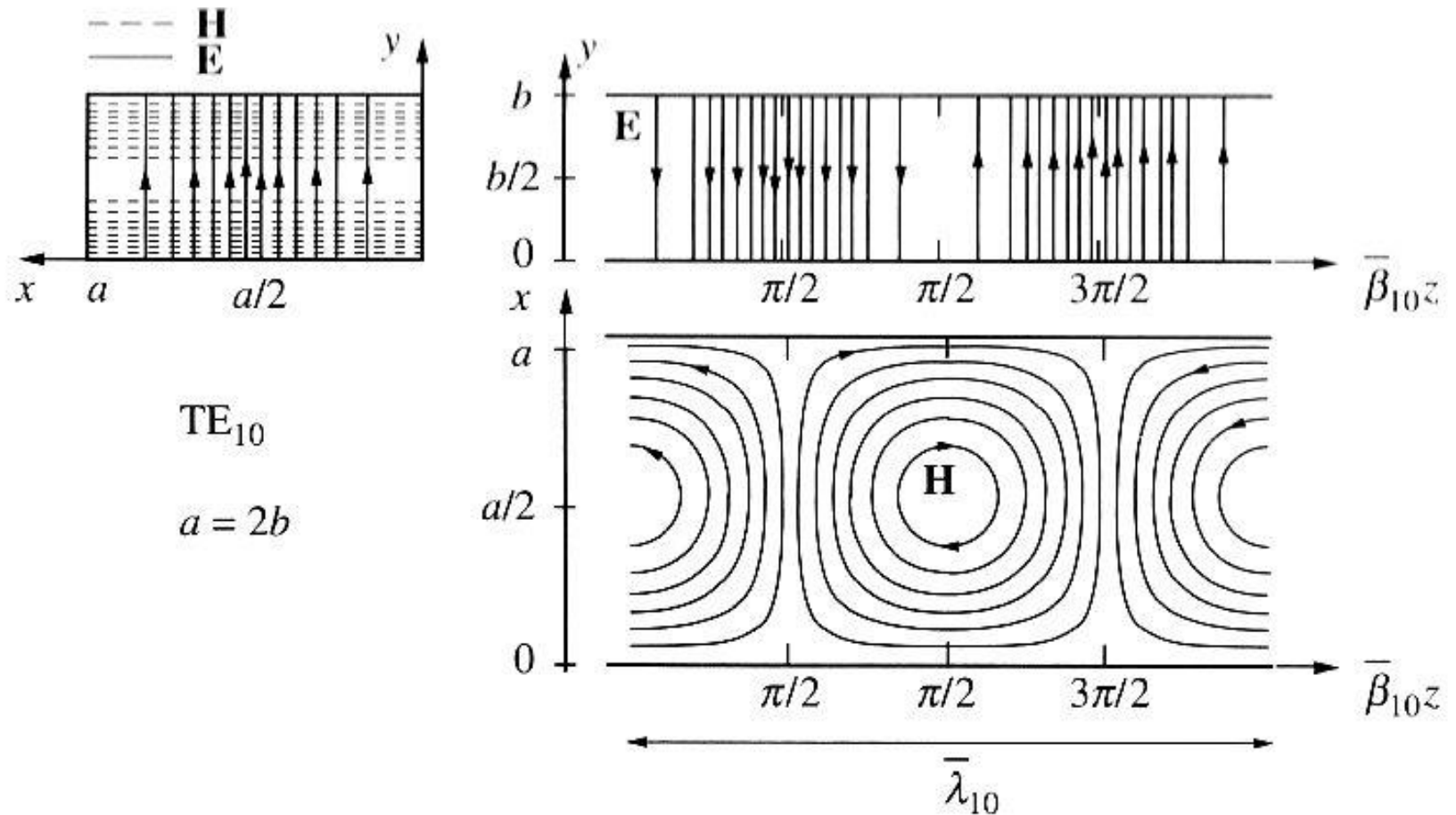
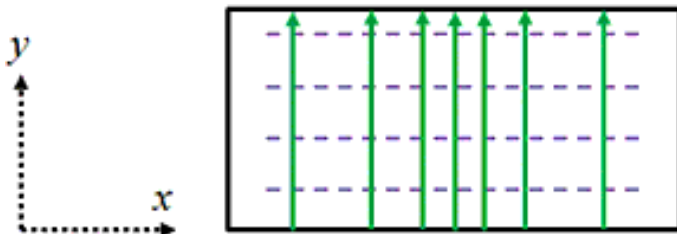
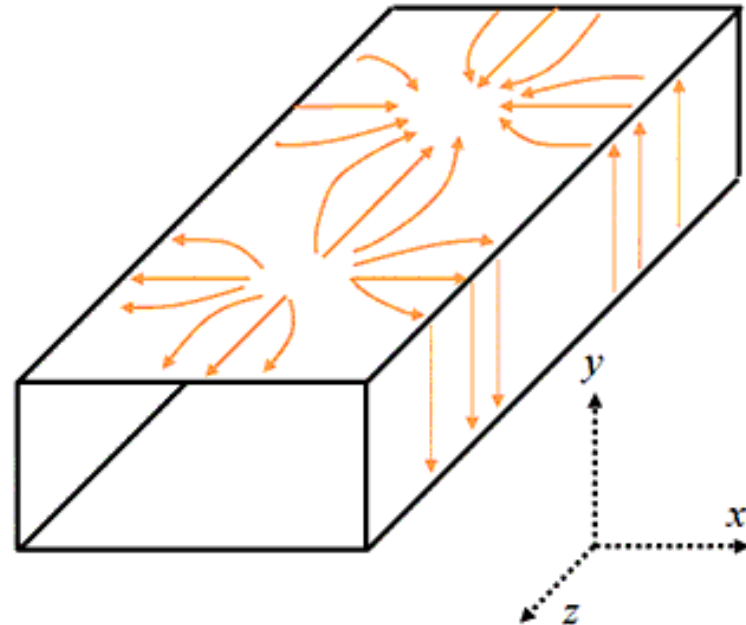
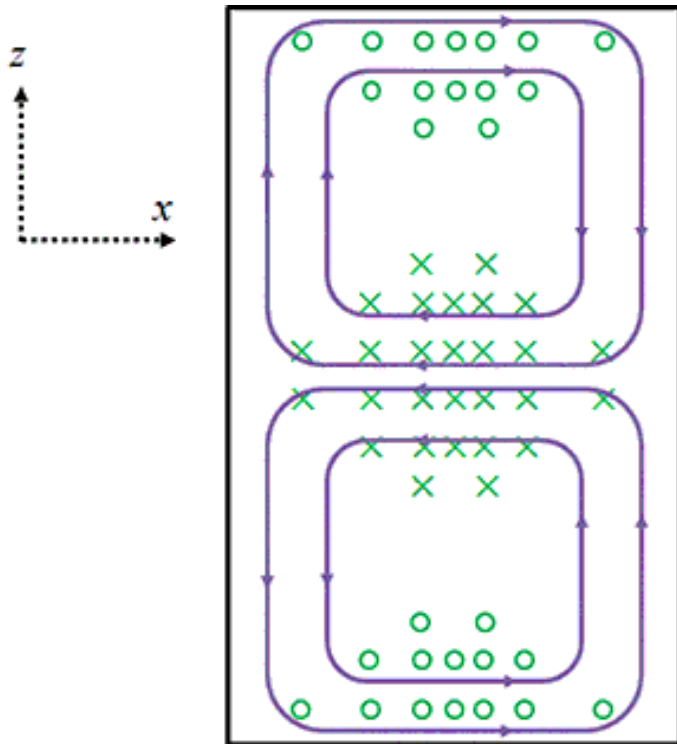


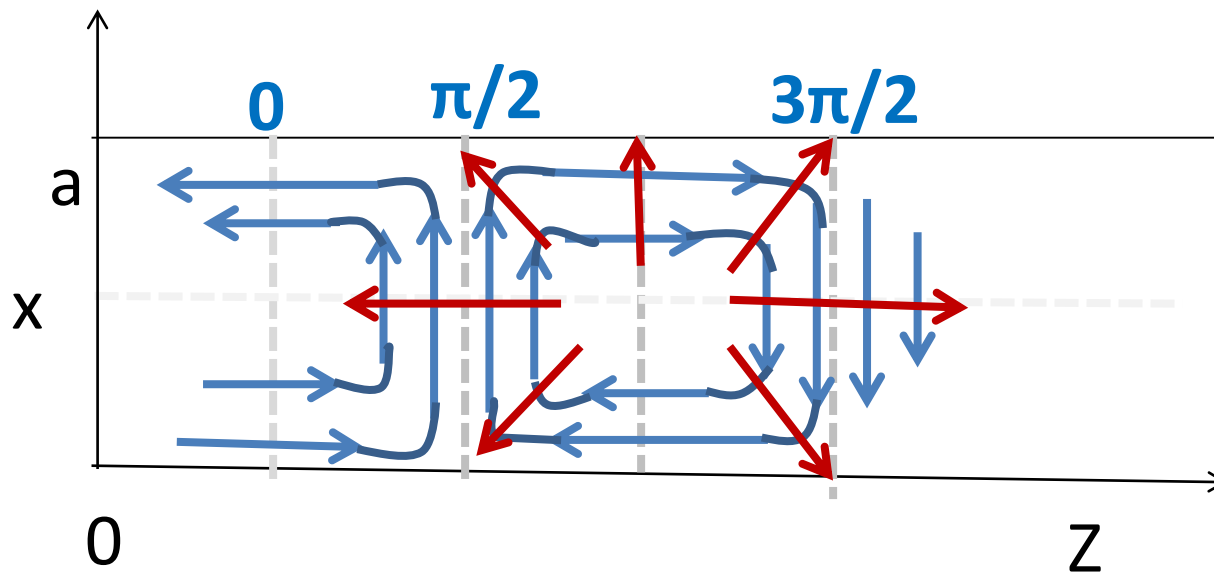
FIGURE 5.2. TE₁₀ mode. The electric and magnetic field configurations for a TE₁₀ mode in a rectangular waveguide.

FIELDS FOR TE₁₀ MODE



FIELDS FOR TE_{10} MODE

Current on Bottom wall



Surface Current $J = \hat{n} \times H_t$

FIELDS FOR TE₁₀ MODE

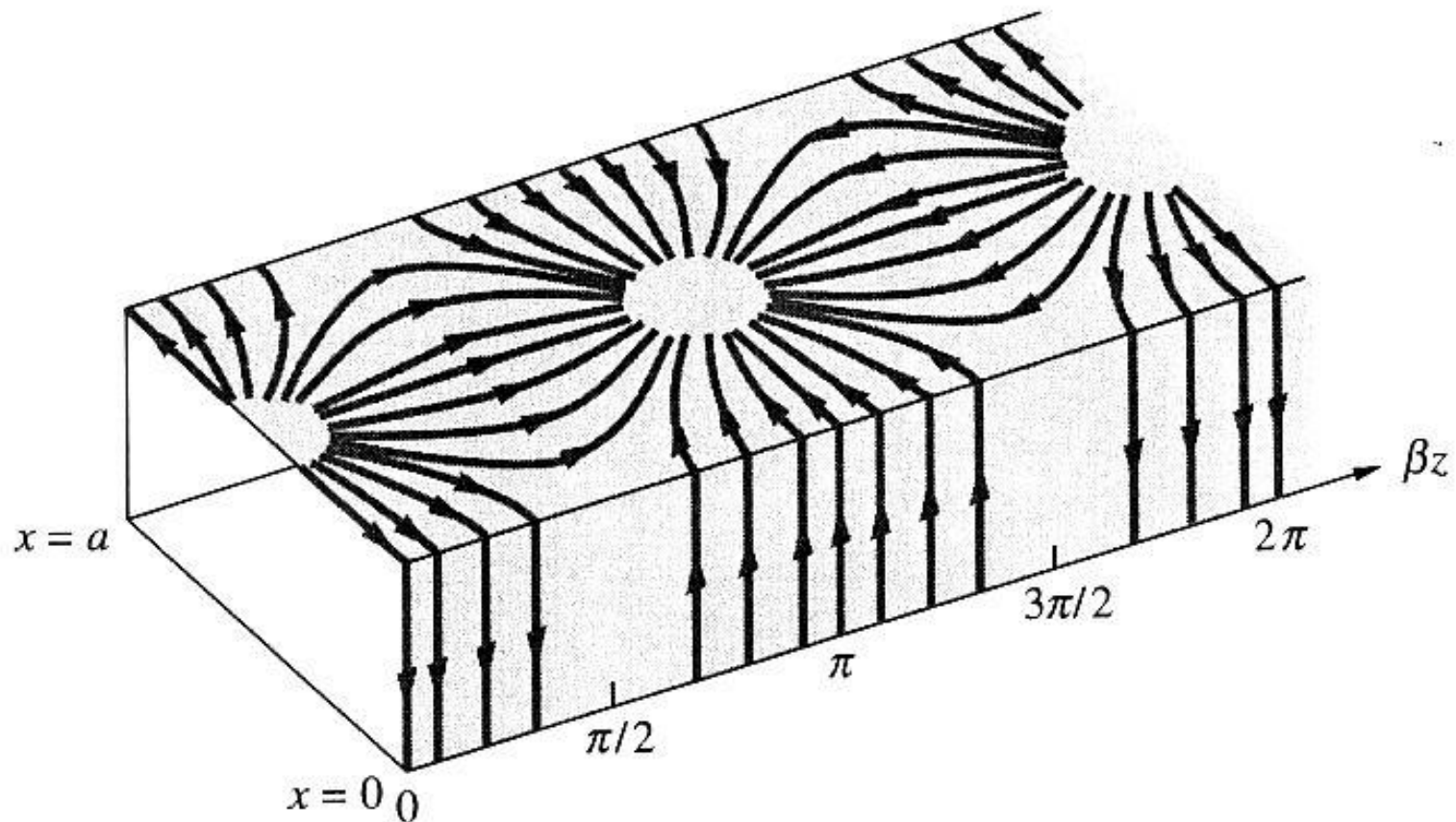
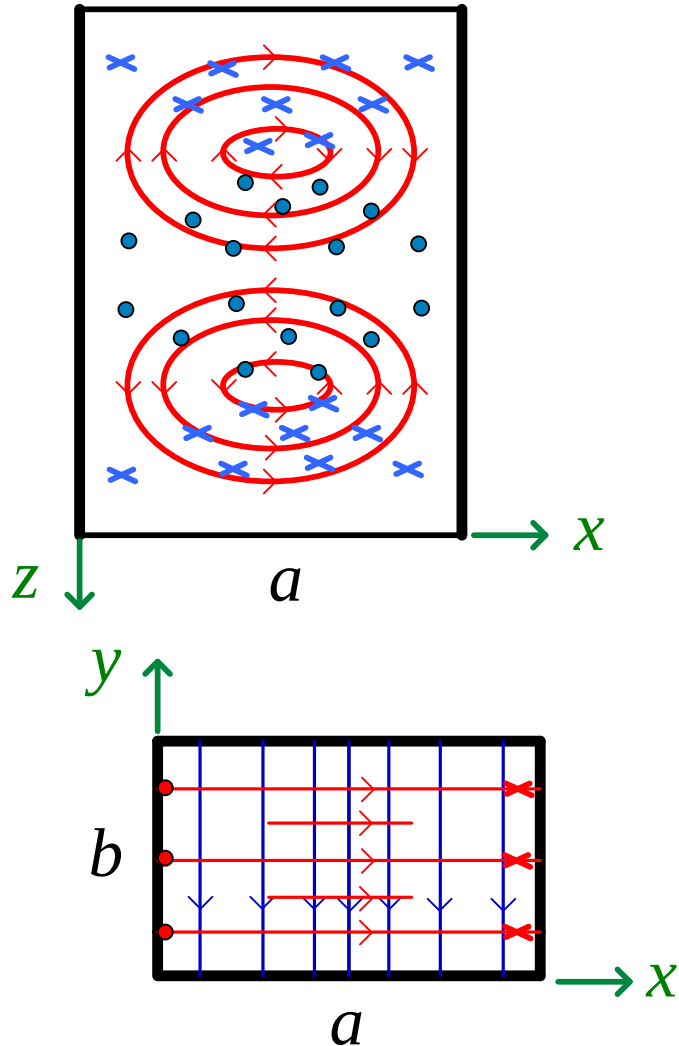


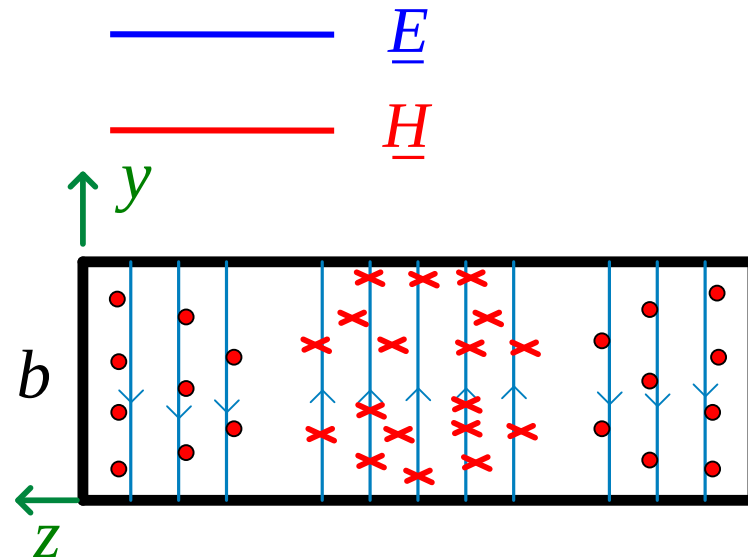
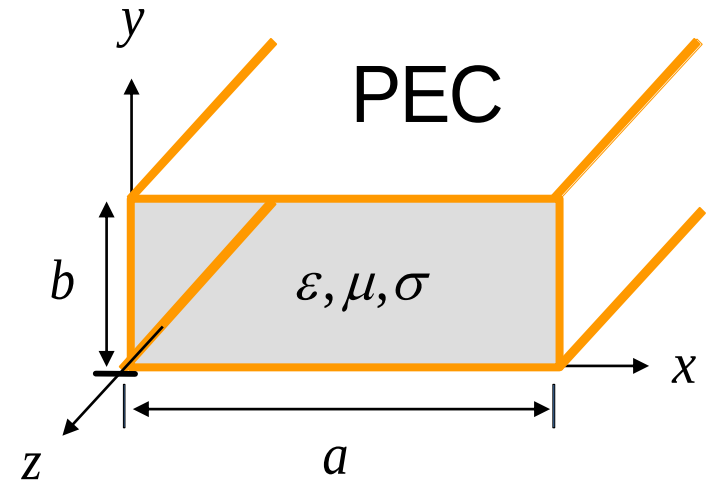
FIGURE 5.7. Surface currents for TE₁₀ mode. Wall currents in a rectangular waveguide for the TE₁₀ mode. [Taken from D. K. Cheng, *Field and Wave Electromagnetics*, Addison-Wesley, Reading, Massachusetts, 1989.]

FIELD PLOTS FOR TE₁₀ MODE

Top view

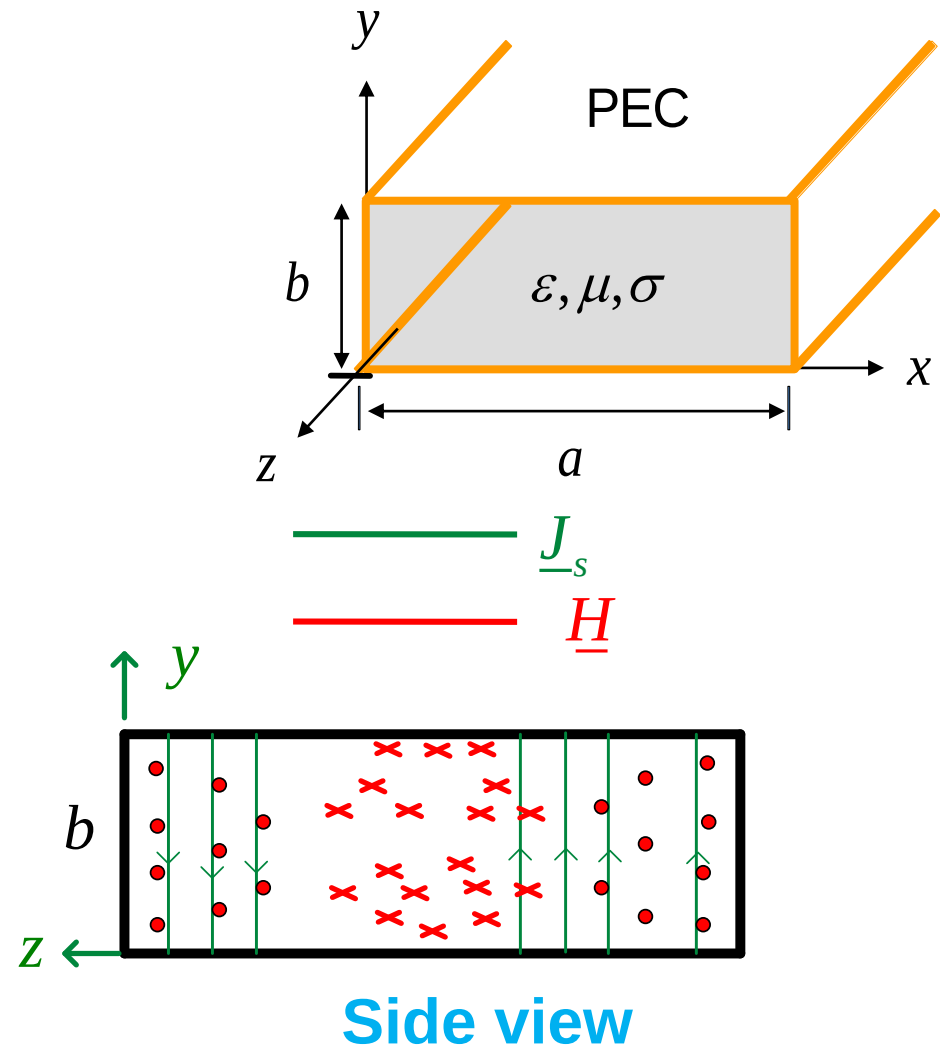
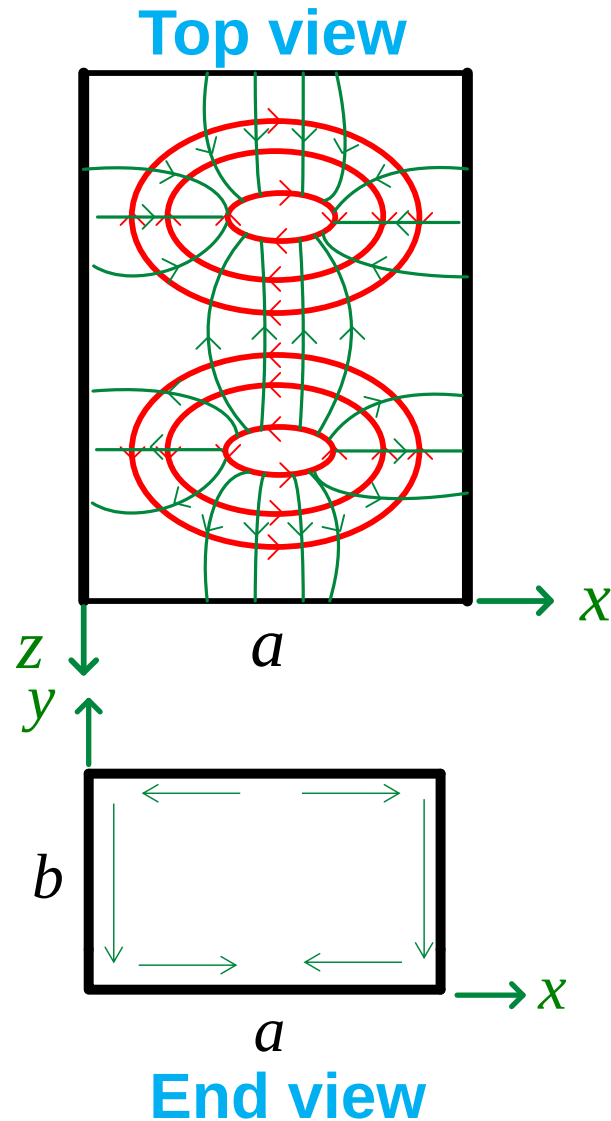


End view



Side view

FIELD PLOTS FOR TE₁₀ MODE (CONT.)



Note: One can cut a narrow z -directed slot in the center of the top wall without disturbing the current.

EXAMPLE:

- $a = 1.07$ cm, $b = 0.43$ cm, $f = 15$ GHz
- Solution: for Teflon $\epsilon_r = 2.08$, $\tan \delta = 0.0004$

$$f_{cmn} = \frac{c}{2\pi\sqrt{\epsilon_r}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

Mode	m	n	f_c (GHz)
TE	1	0	9.72
TE	2	0	19.44
TE	0	1	24.19
TE, TM	1	1	26.07
TE, TM	2	1	31.03

- At 15 GHz

$$\beta = \sqrt{k^2 - \left(\frac{\pi}{a}\right)^2} = 345.1$$

$$\alpha_d = \frac{k^2 \tan \delta}{2\beta} = 0.119 \text{ np / m} = 1.03 \text{ dB / m}$$

$$R_s = \sqrt{\frac{\omega \mu_0}{2\sigma}} = 0.032$$

$$\alpha_c = \frac{R_s}{a^3 b \beta k \eta} (2b\pi^2 + a^3 k^2) = 0.050 \text{ np / m} = 0.434 \text{ dB / m}$$

EXAMPLE:

Consider a length of air-filled copper X-band waveguide, with dimensions $a=2.286\text{cm}$, $b=1.016\text{cm}$ operating at 10GHz . Find the cutoff frequencies of all possible propagating modes.

Solution:

- From the formula for the cut-off frequency

$$f_{cmn} = \frac{v}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

EXAMPLE

Given that $a = 2.286\text{cm}$, $b = 1.016\text{cm}$ and $\sigma = 5.8 \times 10^7\text{S/m}$.
What are the mode and attenuation for 10GHz?

Using this equation to calculate cutoff frequency of each mode

$$f_{cmn} = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

Calculation

TE₁₀

a=2.286mm, b=1.016mm, m=1 and n=0 ,
thus we have

$$f_{c10} = \frac{3 \times 10^8}{2\pi} \sqrt{\left(\frac{\pi}{0.002286} \right)^2} = 6.562 \times 10^9 \text{ GHz}$$

Similarly we can calculate for other modes

EXAMPLE

Mode	f_{cmn}	TE₁₀ TE₂₀ TE₀₁ TE₁₁
TE ₁₀	6.562 GHz	↑ ↑ ↑ ↑
TE ₂₀	13.123GHz	6.562GHz 13.123GHz 14.764GHz 16.156GHz
TE ₀₁	14.764GHz	Frequency 10Ghz is propagating in TE₁₀.mode since this frequency is below the 13.123GHz (TE₂₀) and above 6.561GHz (TE₁₀)
TE ₁₁	16.156GHz	

EXAMPLE (Contd.)

$$\beta = \sqrt{\beta_o - \left(\frac{\pi}{a}\right)^2} = 158.05 m^{-1} \quad R_s = \sqrt{\frac{\omega_o \mu}{2\sigma}} = 0.026 \Omega$$

$$\alpha_c = \frac{R_s}{a^3 b \beta_g \beta_o \eta} (2b \pi^2 + a^3 \beta_o^2) = 0.0125 \quad Np / m$$

or $\alpha_c (dB) = -20 \log e^{-\alpha_c} = 0.11 \quad dB / m$

EXAMPLE

An air-filled 5-by 2-cm waveguide has

$$E_z = 20\sin(40\pi x)\sin(50\pi y)e^{-j\beta z} \text{ V/m}$$

at 15GHz

- What mode is being propagated?
- Find β
- Determine E_y/E_x

EXAMPLE:

Calculate the dimensions of an air-filled rectangular waveguide for which the cutoff frequencies for TM_{11} and TE_{03} modes are both equal to 12 GHz.

EXAMPLE:

$$f_{cmn} = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} = \frac{v}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$f_{c11} = f_{c03} \quad \longrightarrow \quad \frac{v}{2} \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2} = \frac{v}{2} \sqrt{\frac{9}{b^2}}$$

$$\frac{9}{b^2} = \frac{1}{a^2} + \frac{1}{b^2} \quad \longrightarrow \quad a = \frac{b}{\sqrt{8}}$$

$$f_{c03} = \frac{3v}{2b} \quad \longrightarrow \quad b = \frac{3v}{2f_{c03}} = \frac{3 \times 3 \times 10^8}{2 \times 12 \times 10^9} = 3.75 \text{ cm}$$

$$a = 1.32 \text{ cm}, \quad b = 3.75 \text{ cm}$$

SOURCES OF LOSSES

- ❖ Dielectric loss (Losses in the dielectric of filled waveguide)
- ❖ Ohmic loss or conductor loss
$$R_s = \sqrt{\frac{\omega\mu_0}{2\sigma}}$$
- ❖ Radiation/ surface wave losses (Not present in the waveguide).

POWER TRANSMISSION

- ❖ The average Poynting vector for the waveguide fields is

$$\mathcal{P}_{ave} = \frac{1}{2} \operatorname{Re} [E \times H^*] = \frac{1}{2} \operatorname{Re} [E_x H_y^* - E_y H_x^*]$$
$$= \frac{|E_x|^2 + |E_y|^2}{2\eta} \hat{z} \quad [\text{W/m}^2]$$

- ❖ where $\eta = \eta_{TE}$ or η_{TM} depending on the mode

$$P_{ave} = \int \mathcal{P}_{ave} \cdot dS = \int_{x=0}^a \int_{y=0}^b \frac{|E_x|^2 + |E_y|^2}{2\eta} dy dx \quad [\text{W}]$$

ATTENUATION IN LOSSY WAVEGUIDE

- ❖ When dielectric inside guide is lossy, and walls are not perfect conductors, power is lost as it travels along guide.

$$P_{ave} = P_o e^{-2\alpha z}$$

- ❖ The loss power is

$$P_L = -\frac{dP_{ave}}{dz} = 2\alpha P_{ave}$$

- ❖ Where $\alpha = \alpha_c + \alpha_d$ are the attenuation due to ohmic (conduction) and dielectric losses
- ❖ Usually $\alpha_c \gg \alpha_d$

ATTENUATION DUE TO DIELECTRIC LOSS

- But if the line or guide is completely filled with a homogeneous dielectric, the attenuation due to lossy dielectric can be calculated.
- Thus, using the complex dielectric constant allows the complex propagation constant to be written as

$$\gamma^2 + k^2 = k_c^2$$

Derivation

$$\begin{aligned}\gamma &= \alpha_d + j\beta = \sqrt{k_c^2 - k^2} \\ &= \sqrt{k_c^2 - \omega^2 \mu_0 \epsilon_0 \epsilon_r (1 - j \tan \delta)}\end{aligned}$$

$$\begin{aligned}\epsilon_c &= \epsilon - j \frac{\sigma}{\omega} \\ &= \epsilon' - j\epsilon'' - j \frac{\sigma}{\omega} \\ &= \epsilon'_c - j\epsilon''_c \\ &= \epsilon'_c \left(1 - j \frac{\epsilon''_c}{\epsilon'_c} \right) \\ &= \epsilon'_c (1 - j \tan \delta_d) \\ &= \epsilon_0 \epsilon_r (1 - j \tan \delta_d)\end{aligned}$$

ATTENUATION DUE TO DIELECTRIC LOSS

- ❖ In practice, most of dielectric materials have a very small loss ($\tan\delta \ll 1$),
- ❖ So this expression can be simplified by using the first two terms of the Taylor expansion

$$\sqrt{a^2 + x^2} = a \left(1 + \left(\frac{x}{a} \right)^2 \right)^{1/2} \approx a \left(1 + \frac{1}{2} \left(\frac{x}{a} \right)^2 \right) = a + \frac{1}{2} \frac{x^2}{a}$$

for $x \ll a$.

$$\gamma = \sqrt{\underbrace{k_c^2 - k^2}_{a^2} + \underbrace{jk^2 \tan \delta}_{x^2}}$$

$$\gamma = \alpha + j\beta$$

ATTENUATION DUE TO DIELECTRIC LOSS

❖ Then above eq. reduces to

$$\gamma = \sqrt{\underbrace{k_c^2 - k^2}_{a^2} + \underbrace{jk^2 \tan \delta}_{x^2}} \approx \sqrt{k_c^2 - k^2} + \frac{jk^2 \tan \delta}{2\sqrt{k_c^2 - k^2}}$$
$$= \frac{k^2 \tan \delta}{2\beta} + j\beta$$

$$\sqrt{k_c^2 - k^2} = j\beta$$

$$\gamma = \alpha + j\beta$$

❖ For TE or TM wave

$$\alpha_d = \frac{k^2 \tan \delta}{2\beta}$$



$$\alpha_d = \frac{k \tan \delta}{2}$$

❖ For TEM line, $k_c = 0$, $k = \beta$

ATTENUATION DUE TO DIELECTRIC LOSS

- Equation shows that when the loss is small the phase constant β , is unchanged while the attenuation constant due to dielectric loss is given by

$$\alpha_d = \frac{k^2 \tan \delta}{2\beta} = \frac{k^2 \tan \delta}{2k \sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{k \tan \delta}{2 \sqrt{1 - \left(\frac{f_c}{f}\right)^2}} \text{ Np / m (TE or TM waves)}$$

- This result applies to any TE or TM wave, as long as the guide is completely filled with the dielectric. It can also be used for TEM lines, where $k_c=0$, by letting $\beta=k$:

$$\alpha_d = \frac{k \tan \delta}{2} \text{ Np / m (TEM waves)}$$

DOMINANT MODE: TE₁₀ MODE

For this mode we have

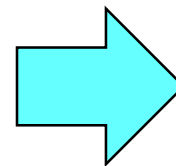
$$m=1, \quad n=0, \quad k_c^{10} = \frac{\pi}{a}$$

Hence we have

$$H_z = H_0 \cos\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

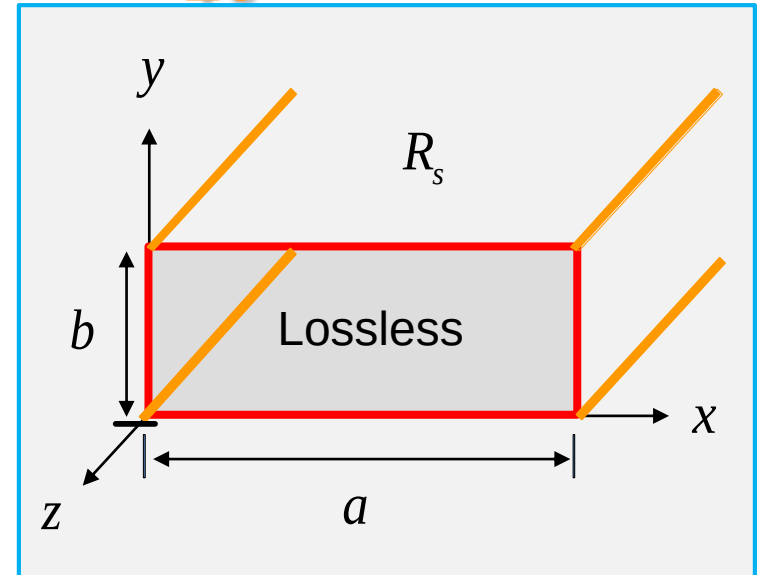
$$H_x = j \frac{k_z a}{\pi} H_0 \sin\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

$$E_y = - \underbrace{\frac{j\omega\mu a}{\pi} H_0}_{E_{10}} \sin\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$



$$E_y = E_{10} \sin\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

$$E_x = E_z = H_y = 0$$



$$\beta = k_z = k_z^{10} = \left(k^2 - \left(\frac{\pi}{a} \right)^2 \right)^{1/2}$$

$$H_0 \equiv \frac{-\pi}{j\omega\mu a} E_{10}$$

Dominant Mode: TE₁₀ Mode (cont.)

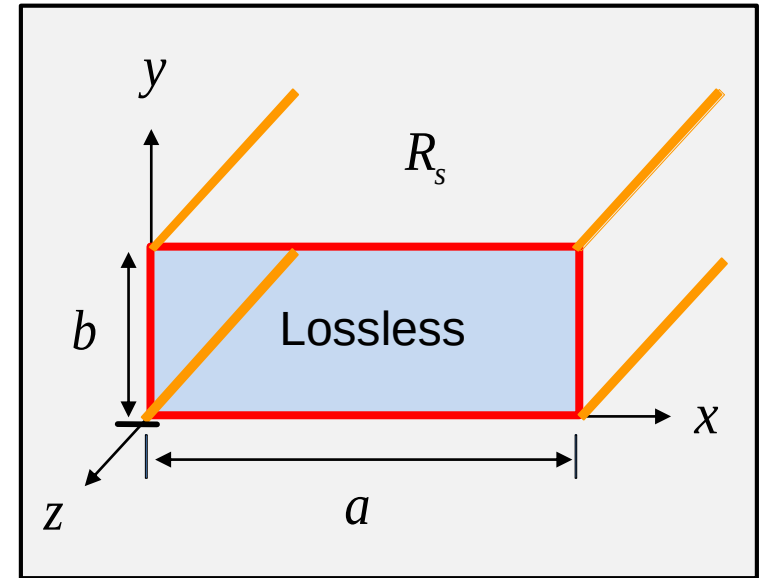
The fields can be put in terms of E_{10} :

$$E_y = E_{10} \sin\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

$$H_x = -\frac{1}{Z_{TE}} E_{10} \sin\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

$$H_z = \left(\frac{-\pi}{j\omega\mu a}\right) E_{10} \cos\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

$$E_x = E_z = H_y = 0$$



$$k_z = k_z^{10} = \left(k^2 - \left(\frac{\pi}{a}\right)^2\right)^{1/2}$$

$$Z_{TE} = \frac{\omega\mu}{k_z}$$

POWER FLOW FOR TE₁₀ MODE

Time-average power flow in the z direction for +z mode:

$$\begin{aligned} P_{10}^+ &= \frac{1}{2} \operatorname{Re} \left\{ \int_0^a \int_0^b (\underline{E} \times \underline{H}^*) \cdot \hat{z} \, dy dx \right\} \\ &= \frac{1}{2} \operatorname{Re} \left\{ \int_0^a \int_0^b -E_y H_x^* \, dy dx \right\} \\ &= \frac{1}{2} \operatorname{Re} \left(\left(\frac{ab}{2} \right) |E_{10}|^2 \left(\frac{k_z}{\omega \mu} \right) e^{-2\alpha z} \right) \end{aligned}$$

Simplifying, we have

$$P_{10}^+ = \left(\frac{ab}{4\omega\mu} \right) \operatorname{Re}\{k_z\} |E_{10}|^2 e^{-2\alpha z}$$

At breakdown:

$$E_{10} = E_c$$

Note:

$$\int_0^a \int_0^b \sin^2 \left(\frac{\pi x}{a} \right) dy dx = \frac{ab}{2}$$

$$E_y = E_{10} \sin \left(\frac{\pi}{a} x \right) e^{-jk_z z}$$

$$H_x = -\frac{1}{Z_{TE}} E_{10} \sin \left(\frac{\pi}{a} x \right) e^{-jk_z z}$$

$$Z_{TE} = \frac{\omega\mu}{k_z}$$

Note:

For a given maximum electric field level (e.g., the breakdown field), the power is increased by increasing the cross-sectional area (ab).

CONDUCTOR ATTENUATION FOR TE₁₀ MODE

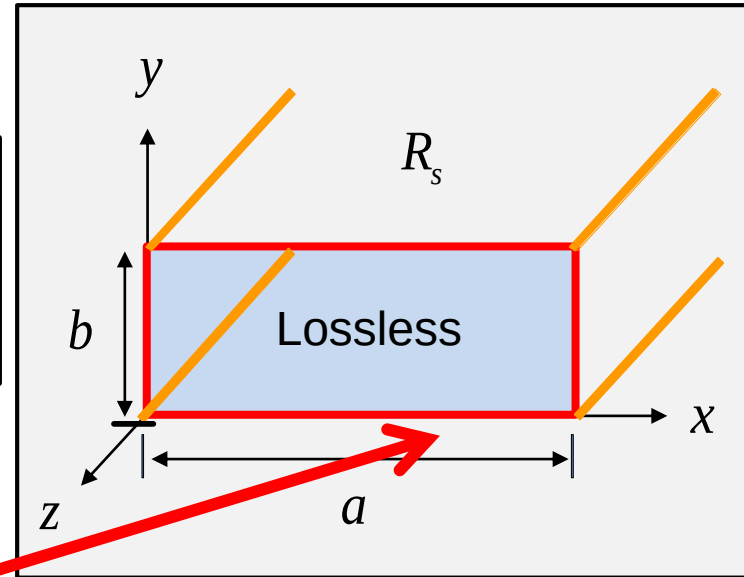
Recall

$$P_0 = P_{10}^+ \Big|_{z=0}$$

$$\alpha_c = \frac{P_l(0)}{2P_0}$$

$$R_s = \sqrt{\frac{\omega\mu_0}{2\sigma}}$$

$$P_l(0) = \frac{R_s}{2} \int_C |\underline{J}_s|^2 d\ell$$



$$C = C^{\text{left}} + C^{\text{right}} + C^{\text{bot}} + C^{\text{top}}$$

$$\underline{J}_s = \underline{\hat{n}} \times \underline{H} \text{ on conductor}$$

The Integration contour C enclose the inside perimeter of the guide walls.

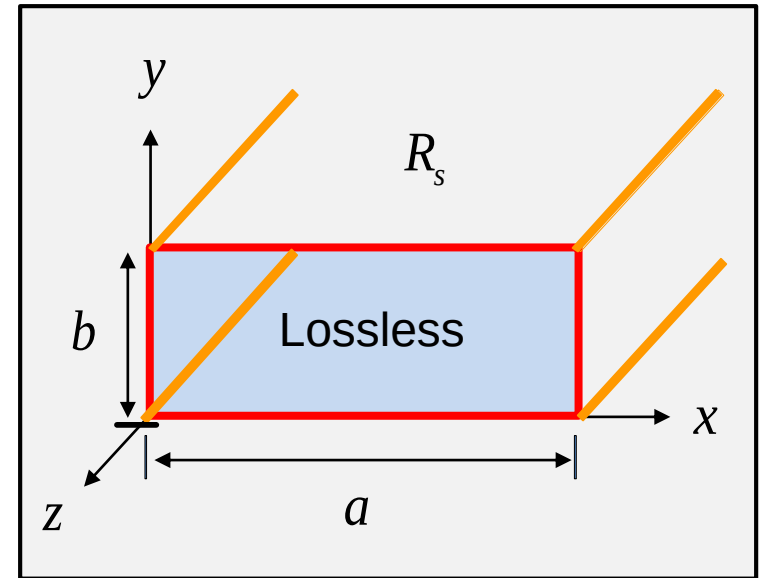
R_s is the wall surface resistance.

CONDUCTOR ATTENUATION FOR TE₁₀ MODE

$$H_z = H_0 \cos\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

$$H_x = j \frac{k_z a}{\pi} H_0 \sin\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

Side walls



$$@ \ x = 0: \quad \underline{J}_s^{\text{left}} = \underline{\hat{x}} \times \underline{H} \Big|_{x=0} = -\underline{\hat{y}} H_z \Big|_{x=0} = -\underline{\hat{y}} H_0 e^{-jk_z z}$$

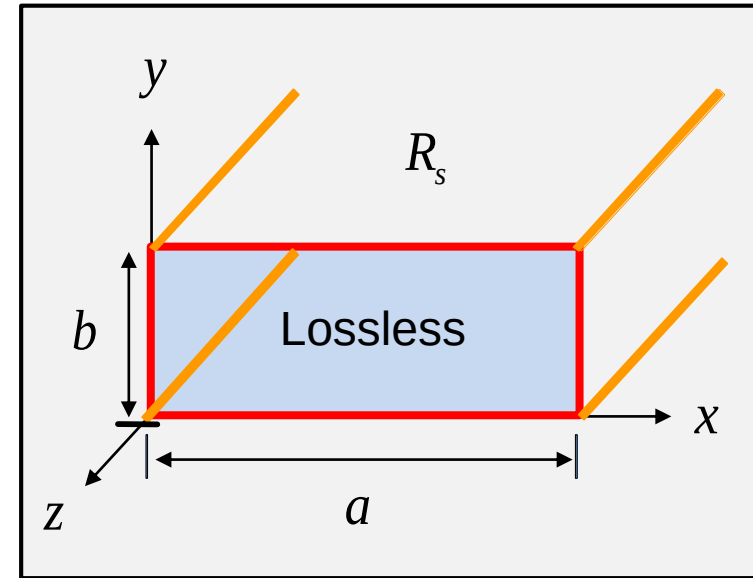
$$@ \ x = a: \quad \underline{J}_s^{\text{right}} = -\underline{\hat{x}} \times \underline{H} \Big|_{x=a} = \underline{\hat{y}} H_z \Big|_{x=a} = -\underline{\hat{y}} H_0 e^{-jk_z z}$$

$$\text{Hence:} \quad J_{sy}^{\text{left}} = J_{sy}^{\text{right}} = -H_0 e^{-jk_z z}$$

CONDUCTOR ATTENUATION FOR TE₁₀ MODE (CONT.)

$$H_z = H_0 \cos\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$

$$H_x = j\frac{k_z a}{\pi} H_0 \sin\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$



Bottom walls

$$@ \ y = 0: \quad \underline{J}_s^{\text{bot}} = \underline{\hat{y}} \times \underline{H} \Big|_{y=0}$$

$$= \underline{\hat{y}} \times \underline{\hat{x}} \underline{H}_x \Big|_{y=0} + \underline{\hat{y}} \times \underline{\hat{z}} \underline{H}_z \Big|_{y=0}$$

$$= -\underline{\hat{z}} \underline{H}_x \Big|_{y=0} + \underline{\hat{x}} \underline{H}_z \Big|_{y=0}$$

$$= \underline{\hat{z}} J_{sz}^{\text{bot}} + \underline{\hat{x}} J_{sx}^{\text{bot}}$$

$$J_{sx}^{\text{bot}} = H_0 \cos\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$

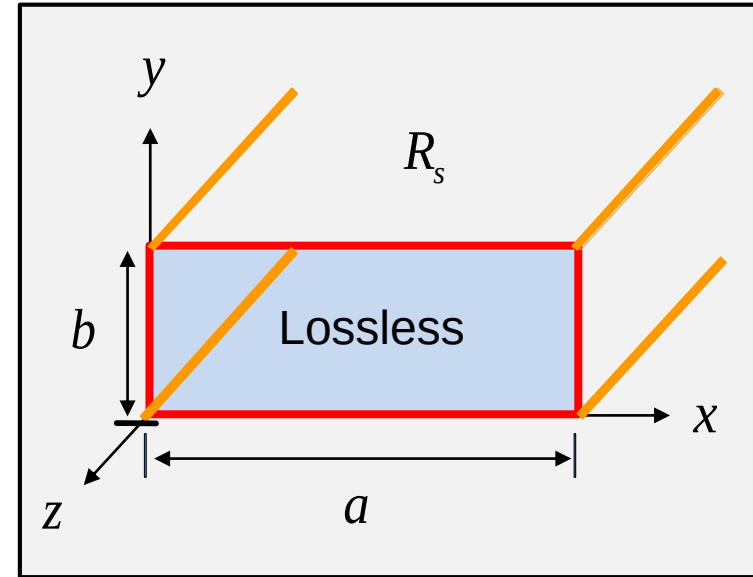
$$J_{sz}^{\text{bot}} = -j\frac{k_z a}{\pi} H_0 \sin\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$

CONDUCTOR ATTENUATION FOR TE₁₀ MODE (CONT.)

$$H_z = H_0 \cos\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$

$$H_x = j\frac{k_z a}{\pi} H_0 \sin\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$

Top walls



$$@ \quad y = b : \quad \underline{J}_s^{\text{top}} = -\underline{\hat{y}} \times \underline{H} \Big|_{y=b}$$

$$J_{sx}^{\text{top}} = -H_0 \cos\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$

$$J_{sz}^{\text{top}} = j\frac{k_z a}{\pi} H_0 \sin\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$

$$= -\underline{\hat{y}} \times \underline{\hat{x}} \underline{H}_x \Big|_{y=b} - \underline{\hat{y}} \times \underline{\hat{z}} \underline{H}_z \Big|_{y=b}$$

$$= \underline{\hat{z}} \underline{H}_x \Big|_{y=b} - \underline{\hat{x}} \underline{H}_z \Big|_{y=b}$$

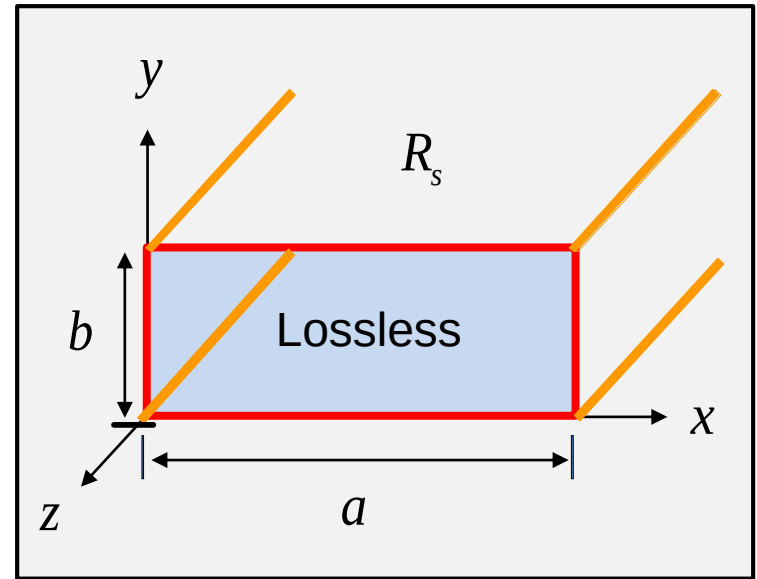
$$= \underline{\hat{z}} J_{sz}^{\text{top}} + \underline{\hat{x}} J_{sx}^{\text{top}}$$

CONDUCTOR ATTENUATION FOR TE₁₀ MODE (CONT.)

$$H_z = H_0 \cos\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$
$$H_x = j\frac{k_z a}{\pi}H_0 \sin\left(\frac{\pi}{a}x\right)e^{-jk_z z}$$

Hence:

$$\underline{J}_s^{\text{top}} = -\underline{J}_s^{\text{bot}}$$



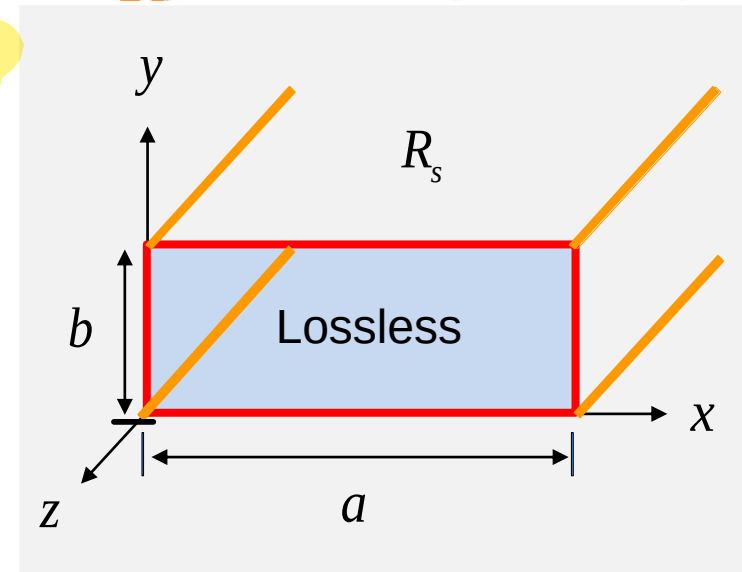
$$J_{sx}^{\text{bot}} = H_0 \cos\left(\frac{\pi}{a}x\right)e^{-jk_z z} = -J_{sx}^{\text{top}}$$

$$J_{sz}^{\text{bot}} = -j\frac{k_z a}{\pi}H_0 \sin\left(\frac{\pi}{a}x\right)e^{-jk_z z} = -J_{sz}^{\text{top}}$$

CONDUCTOR ATTENUATION FOR TE₁₀ MODE (CONT.)

We then have: **Derivation**

$$\begin{aligned}
 P_l(0) &= 2 \left(\frac{R_s}{2} \int_0^b \left| \underline{J}_s^{\text{left}} \right|^2 dy + \frac{R_s}{2} \int_0^a \left| \underline{J}_s^{\text{bot}} \right|^2 dx \right) \\
 &= R_s \int_0^b \left| J_{sy}^{\text{left}} \right|^2 dy + R_s \int_0^a \left(\left| J_{sx}^{\text{bot}} \right|^2 + \left| J_{sz}^{\text{bot}} \right|^2 \right) dx \\
 &= R_s \int_0^b \left| -H_0 \right|^2 dy + R_s \int_0^a \left(\left| H_0 \cos \left(\frac{\pi}{a} x \right) \right|^2 + \left| -j \frac{k_z a}{\pi} H_0 \sin \left(\frac{\pi}{a} x \right) \right|^2 \right) dx \\
 &= R_s |H_0|^2 \left(\int_0^b dy + \int_0^a \cos^2 \left(\frac{\pi}{a} x \right) dx + \left(\frac{|k_z| a}{\pi} \right)^2 \int_0^a \sin^2 \left(\frac{\pi}{a} x \right) dx \right) \\
 &= R_s |H_0|^2 \left(b + \frac{a}{2} + \frac{|k_z|^2 a^2}{\pi^2} \frac{a}{2} \right)
 \end{aligned}$$



$$P_l(0) = R_s |H_0|^2 \left(b + \frac{a}{2} + \frac{|k_z|^2 a^3}{2\pi^2} \right)$$

ATTENUATION FOR TE₁₀ MODE (CONT.)

Assume $f > f_c$ $k_z = \gamma \approx \beta$

(The wavenumber is taken as that of a guide with perfect walls.)

$$P_l(0) = R_s |H_0|^2 \left(b + \frac{a}{2} + \frac{\beta^2 a^3}{2\pi^2} \right)$$

$$H_0 = \frac{-\pi}{j\omega\mu a} E_{10}$$

$$P_0 = \left(\frac{ab}{4\omega\mu} \right) \beta |E_{10}|^2$$

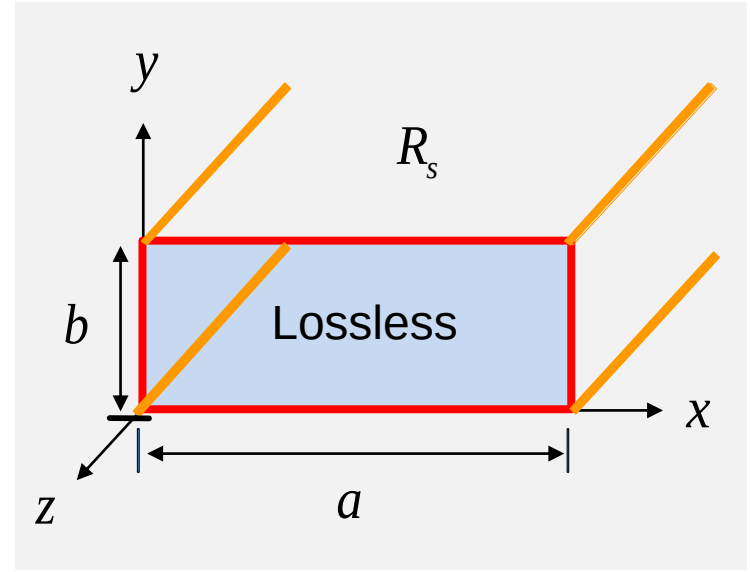
$$\alpha_c = \frac{P_l(0)}{2P_0}$$

Simplify, using

$$\beta^2 = k^2 - k_c^2 \quad k_c^{10} = \frac{\pi}{a}$$

Final result:

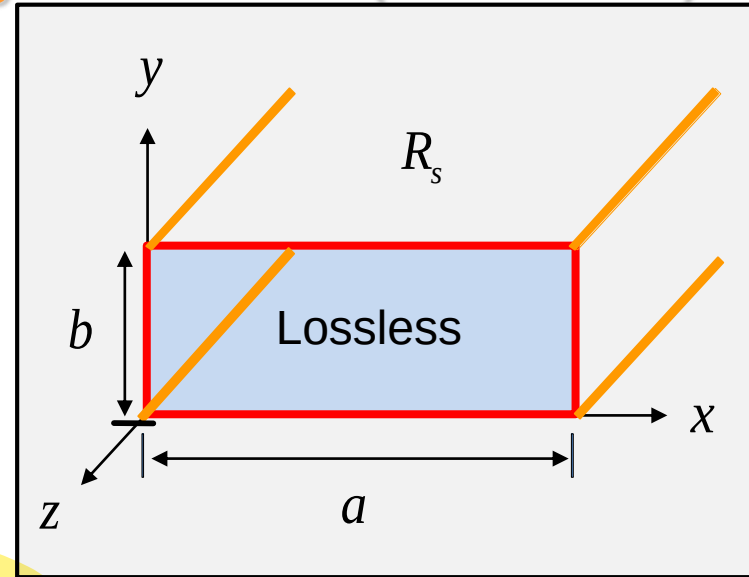
$$\alpha_c = \frac{R_s}{a^3 b \beta (k\eta)} (2b\pi^2 + a^3 k^2) \quad [\text{np/m}]$$



ATTENUATION FOR TE₁₀ MODE (CONT.)

Final Formulas

Two alternative forms for the final result:



$$\alpha_c = \frac{R_s}{a^3 b \beta(k\eta)} (2b\pi^2 + a^3 k^2) \quad [\text{np/m}]$$

$$\alpha_c = \frac{R_s}{b\eta} \frac{1}{\sqrt{1 - (f_c / f)^2}} \left(1 + \frac{2b}{a} \left(\frac{f_c}{f} \right)^2 \right) \quad [\text{np/m}]$$

ATTENUATION FOR TE₁₀

Dielectric loss

❖ Dielectric attenuation, Np/m

$$\alpha_d = \frac{k \tan \delta}{2 \sqrt{1 - \left(\frac{f_c}{f} \right)^2}} \text{ Np / m (TE or TM waves)}$$

❖ Conductor attenuation, Np/m

$$\alpha_c = - \frac{2R_s}{b\eta' \sqrt{1 - \left(\frac{f_{c,10}}{f} \right)^2}} \left(0.5 + \frac{b}{a} \left(\frac{f_{c,10}}{f} \right)^2 \right)$$

$$\epsilon_c = \begin{cases} \epsilon - j \frac{\sigma}{\omega} \\ \epsilon' - j\epsilon'' - j \frac{\sigma}{\omega} \\ \epsilon'_c - j\epsilon''_c \\ \epsilon'_c \left(1 - j \frac{\epsilon''_c}{\epsilon'_c} \right) \\ \epsilon'_c (1 - j \tan \delta_d) \\ \epsilon_0 \epsilon_r (1 - j \tan \delta_d) \end{cases}$$

ATTENUATION FOR TE₁₀ MODE (CONT.)

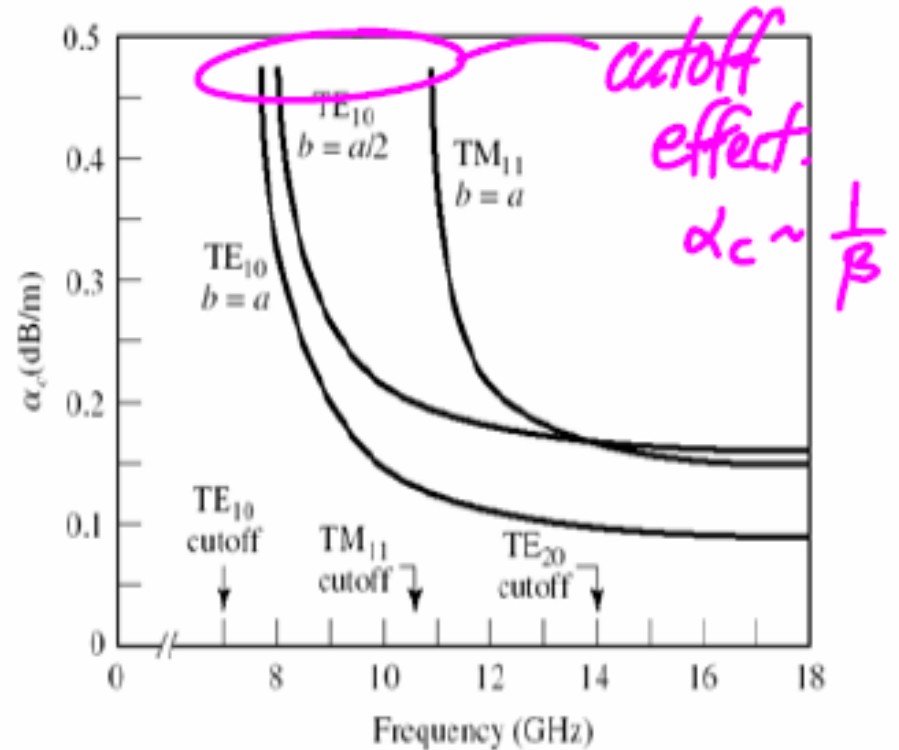
Brass X-band air-filled waveguide

$$(\sigma \approx 2.6 \times 10^7 \text{ [S/m]})$$

X band: 8–12 [GHz]

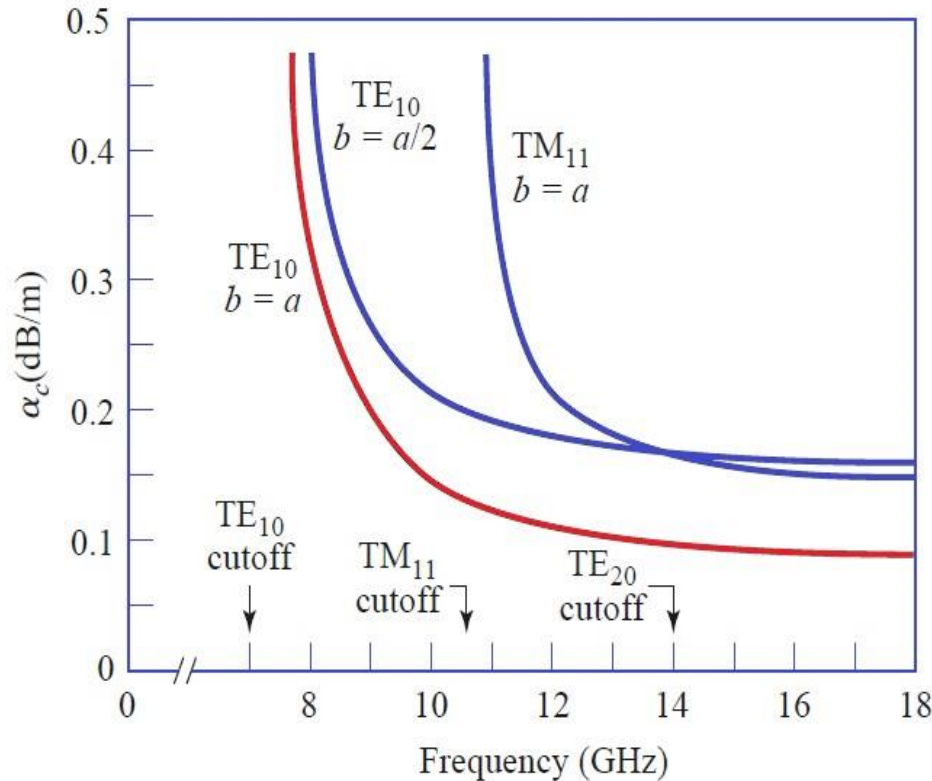
(See the table on the next slide.)

$$a = 2.0 \text{ cm}$$



(from the Pozar book)

ATTENUATION FOR TE₁₀ MODE (CONT.)



Variation of attenuation constant for $a = 20$ mm.

Near Cutoff frequency $\alpha_c \propto \frac{1}{\beta}$

Fast wave: $\lambda_g = \frac{2\pi}{\beta} > \frac{2\pi}{k}$

Cutoff frequency for TE₁₀ mode:

$$f_{c10} = \frac{1}{2a\sqrt{\mu\epsilon}}$$

• Propagation constant:

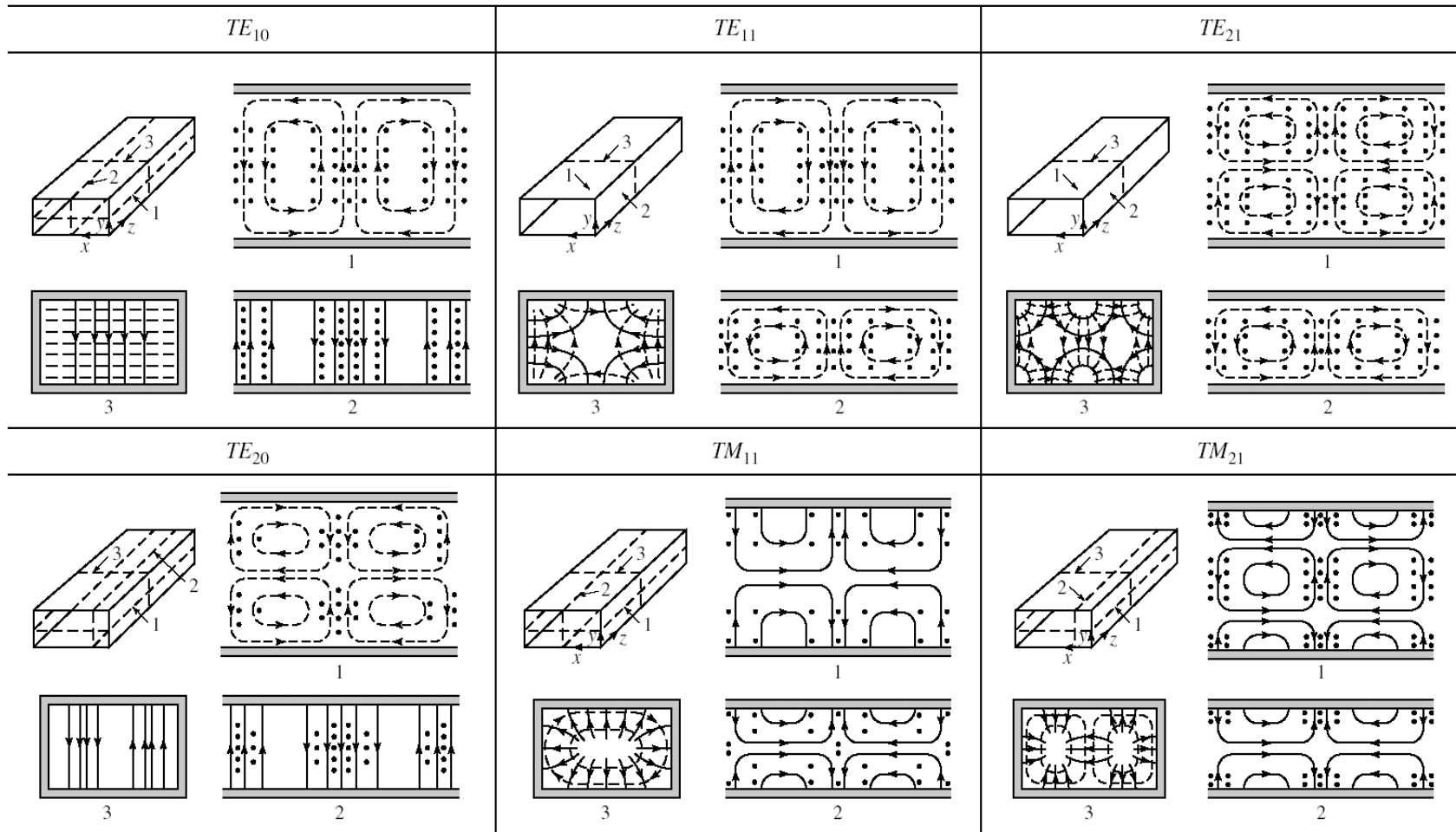
$$\beta = k_z = k_z^{10} = \left(k^2 - \left(\frac{\pi}{a} \right)^2 \right)^{1/2}$$

Power loss in the guide:

$$\alpha_c = \frac{R_s}{a^3 b \beta (k\eta)} (2b\pi^2 + a^3 k^2) \quad [\text{np/m}]$$

$$v_p = \frac{\omega}{\beta} > \frac{\omega}{k}$$

FIELD LINES FOR SOME OF THE LOWER ORDER MODES OF A RECTANGULAR WAVEGUIDE.



MODES IN AN X-BAND WAVEGUIDE

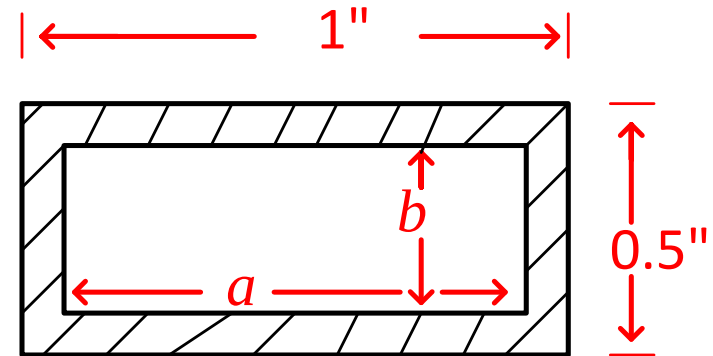
$$a = 2.29 \text{ cm (0.90 in)}$$

$$b = 1.02 \text{ cm (0.40 in)}$$

“Standard X-band waveguide” (WR90)

X band: 8–12 [GHz]

Mode	f_c [GHz]
TE_{10}	6.55
TE_{20}	13.10
TE_{01}	14.71
TE_{11}	16.10
TM_{11}	16.10
TE_{30}	19.65
TE_{21}	19.69
TM_{21}	19.69

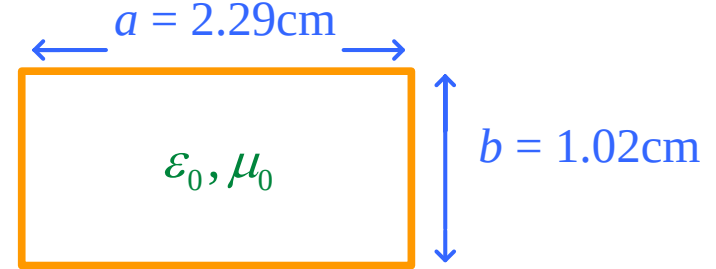


50 mil (0.05") thickness

$$f_c^{mn} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

EXAMPLE: X-BAND WAVEGUIDE

Determine β , α , and λ_g (as appropriate) at 10 GHz and 6 GHz for the TE₁₀ mode in a lossless air-filled X-band waveguide.



@ 10 GHz

$$\beta = \sqrt{\omega^2 \mu \epsilon - \left(\frac{\pi}{a}\right)^2} = \sqrt{\left(\frac{2\pi 10^{10}}{2.99792458 \times 10^8}\right)^2 - \left(\frac{\pi}{0.0229}\right)^2}$$

$$\lambda_g = \frac{2\pi}{\beta} = \frac{2\pi}{158.25} = 0.0397$$

$$\beta = 158.25 \text{ [rad/m]}$$

$$\lambda_g = 3.97 \text{ [cm]}$$

$$\text{Lossless: } k = \frac{2\pi}{\lambda'} = \frac{2\pi}{v/f} = \frac{2\pi f}{v} = \frac{\omega}{v} = \omega \sqrt{\mu \epsilon} = 2\pi f \sqrt{\mu \epsilon}$$

EXAMPLE: X-BAND WAVEGUIDE (CONT.)

@ 6 GHz

$$\begin{aligned} k_z &= \left(\omega^2 \mu \epsilon - \left(\frac{\pi}{a} \right)^2 \right)^{1/2} = \left(\left(\frac{2\pi 6 \times 10^9}{2.99792458 \times 10^8} \right)^2 - \left(\frac{\pi}{0.0229} \right)^2 \right)^{1/2} \\ &= -j \sqrt{\left(\frac{\pi}{0.0229} \right)^2 - \left(\frac{2\pi 6 \times 10^9}{2.99792458 \times 10^8} \right)^2} \\ &= -j 55.04 \text{ [1/m]} \end{aligned}$$

$$\lambda_g = \frac{2\pi}{\beta}$$

$$\begin{aligned} \alpha &= 55.04 \text{ [np/m]} \\ &= 478.08 \text{ [dB/m]} \end{aligned}$$

Evanescent mode: $\beta = 0$; λ_g is not defined!

EXAMPLE: WAVEGUIDE (CONT.)

Find the single-mode operating frequency region for air-filled X-band waveguide.

Standard X-band* waveguide:

$$a = 0.900 \text{ inches (2.286 cm)}$$

$$b = 0.400 \text{ inches (1.016 cm)}$$

Use $f_c^{(1,0)} = \frac{c}{2a}$

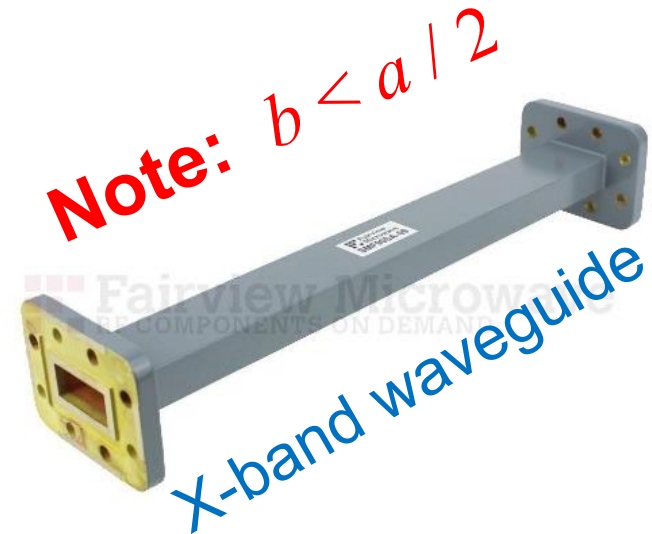
Hence, we have:

$$f_c^{(1,0)} = 6.56 \text{ [GHz]}$$

$$f_c^{(2,0)} = 13.11 \text{ [GHz]}$$

$$6.56 < f < 13.11 \text{ [GHz]}$$

* X-band: from 8.0 to 12 GHz.



EXAMPLE: WAVEGUIDE (CONT.)

- ❖ Find the phase constant of the TE_{10} mode at 9.00 GHz.
- ❖ Find the attenuation in dB/m at 5.00 GHz

Recall: $f_c^{(1,0)} = 6.56$ [GHz]

$$\beta = \sqrt{k^2 - \left(\frac{\pi}{a}\right)^2}, \quad f > f_c$$

$$\alpha = \sqrt{\left(\frac{\pi}{a}\right)^2 - k^2}, \quad f < f_c$$

$$k_c = \pi / a = 137.43 \text{ [rad/m]}$$

$$k = k_0 = \omega \sqrt{\mu_0 \epsilon_0} = 2\pi f / c = 2\pi / \lambda_0$$

At 9.0 GHz: $k = 188.62$ [rad/m]

At 5.0 GHz: $k = 104.79$ [rad/m]



At 9.00 GHz: $\beta = 129.13$ [rad/m]

At 5.00 GHz: $\alpha = 88.91$ [nepers/m]

EXAMPLE: WAVEGUIDE (CONT.)

At 5.0 GHz:

$$\alpha = 88.91 \text{ [nepers/m]}$$

Recall:

$$\text{dB/m} = 8.68589 \alpha$$

$$\text{Attenuation} = 772 \text{ dB/m}$$

This is a very rapid attenuation!



WAVEGUIDE COMPONENTS



Straight sections



Flexible
waveguides



Waveguide bends



Waveguide
adapters



Waveguide
couplers



Waveguide
terminations

WAVEGUIDE MODES IN TRANSMISSION LINES COMPONENTS

- ❖ A transmission line normally operates in the TEM_z mode, where the two conductors have equal and opposite currents.
- ❖ At high frequencies, waveguide modes can also propagate on transmission lines.
- ❖ This is undesirable, and limits the high-frequency range of operation for the transmission line.

IMPOSSIBILITY OF TEM WAVE IN WAVE GUIDES

- Suppose a TEM wave is assumed to exist within a hollow guide of any shape.
- Then lines of H must lie entirely in the transverse plane.
- Also in a nonmagnetic material,

$$\nabla \cdot \mathbf{H} = 0$$

which requires that the lines of H be *closed* loops.

- Therefore, if a TEM wave exists in the guide, the lines of H will be closed loops in plane perpendicular to the axis.

IMPOSSIBILITY OF TEM WAVE IN WAVE GUIDES(CONTD.)

- Now by Maxwell's first equation the magnetomotive force around each of these closed loops must equal the axial current (conduction or displacement) through the loop.

$$\oint_L H \cdot dS = \int (\dot{D} + J) \cdot da = \int (J_d + J_c) \cdot da$$

- However, for a hollow wave guide having no inner conductor, axial conduction($J_c=0$)current is 0

IMPOSSIBILITY OF TEM WAVE IN WAVE GUIDES(CONTD.)

- The axial current must be a displacement current (J_d).

$$J_d = \dot{D}_z = \epsilon \frac{\partial E_z}{\partial t}$$

- But an axial displacement current requires an axial component of E , something not present in a TEM wave.
- Therefore the TEM wave cannot exist in a single-conductor wave guide